



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
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NBA Accredited (UG - 6 Years)

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year: 2026-27

Principles of Digital Communication Systems (EC353AI)

Laboratory Manual and Observation Book

(Autonomous Scheme 2022)



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RV COLLEGE OF ENGINEERING®, Bengaluru-59

(Autonomous Institution affiliated to VTU, Belagavi)

**Department of Electronics and Communication
Engineering**

Laboratory Certificate

This is to certify that Mr. / Ms _____
_____ has satisfactorily completed the course of Experiments
in Principles of Digital Communication Systems lab (EC353AI) course prescribed by
the Department during the year _____

USN No.: _____ Semester: _____

Marks	
Maximum	Obtained
50	

Marks in words

Signature of the staff in-charge

Head of the Department

Date:

RV COLLEGE OF ENGINEERING®, Bengaluru-59

(Autonomous Institution affiliated to VTU, Belagavi)

Department of Electronics and Communication Engineering

SCHEME OF CONDUCT AND EVALUATION

CLASS: 5th SEMESTER

YEAR: 2026-27

CIE MARKS (Max): 50

SEE MARKS (Max): 50

SEE: 03 Hrs

Expt No	Title	Duration in Hrs	Max Marks
1	M-PSK transceiver Design using Labview	2	10
2	M-QAM transceiver Design using Labview	2	10
3	ISI with PAM modulation technique using MATLAB	2	10
4	Equalization with QAM Modulation using Labview	2	10
5	Transmitter and Receiver Implementation of Direct Sequence Spread Spectrum and Frequency Hopped Spread Spectrum using MATLAB	2	10
6	Symbol Timing Estimation using MATLAB	2	10
7	Frequency selective and non-selective fading channel simulation using MATLAB	2	10
8	Carrier Phase Estimation: Decision-Directed and Non-Decision-Directed Loops using MATLAB	2	10
9	M-PSK transceiver Design using USRP 2920 Software Defined Radio	2	10
10	M-QAM transceiver Design using USRP 2920 Software Defined Radio	2	10
11	M-FSK transceiver Design using USRP 2920 Software Defined Radio	2	10
12	Adaptive Equalization: Zero-Forcing (ZFE), MSE/LMS and Decision-Feedback Equalizer (DFE)	2	10
Lab Record Marks		20	
Lab EL		20	
Lab CIE		10	
Total Lab CIE Marks		50	

Sl.No	Criteria	Excellent	Good	Average	Max Score
Data sheet					
A	Problem statement	9-10	6-8	1-5	10
B	Design & specifications	9-10	6-8	1-5	10
C	Expected output	9-10	6-8	1-5	10
Record					
D	Simulation/ Conduction of the experiment	14-15	11-13	1-10	15
E	Analysis of the result.	14-15	11-13	1-10	15
Viva					40
Total					100
Scale down to 10 marks for each experiment					

Rubrics for Lab Record evaluation

Course Outcomes	
1	Utilize mathematical tools for signal processing and system modeling to design and optimize communication systems for specific applications.
2	Apply knowledge of digital communication theory to design and analyze communication systems for specific applications, considering factors such as bandwidth, data rate, power efficiency, and error performance.
3	Analyze the performance of digital communication systems through mathematical analysis and simulation, assessing metrics such as bit error rate, signal-to-noise ratio, and bandwidth efficiency.
4	Test and validate digital formatting schemes and block codes under noisy channel conditions to estimate the performance in practical communication systems.

EXPERIMENT 1

M-PSK Transceiver Design using Labview Software

Aim: To implement an M-PSK transceiver implementation using Labview software

Tools required: Desktop or Laptop with Labview software

Theory:

Phase-shift keying (PSK) in a digital transmission refers to a type of angle modulation in which the phase of the carrier is discretely varied to represent data being transmitted either in relation to a reference phase, or to the phase of the immediately preceding signal element. For example, when encoding bits, the phase shift could be 0° for encoding a 0 and 180° for encoding a 1, or the phase shift could be -90° for 0 and $+90^\circ$ for a 1, thus making the representations for 0 and 1 a total of 180° apart. In PSK systems designed so that the carrier can assume only two different phase angles, each change of phase carries one bit of information, that is, the bit rate equals the modulation rate. If you increase the number of recognizable phase angles to four, then 2 bits of information can be encoded into each signal element; likewise, eight phase angles can encode 3 bits in each signal element.

Procedure:

1. The basic block diagram of M-PSK transceiver in Labview is shown in figure 1.1.
2. The corresponding front panel in Labview is shown in figure 1.2, 1.3 and 1.4 for $M=4, 8$ and 16 respectively.
3. Vary the M-levels in front panel (Ex $M=2, 4, 8, 16$ etc) and observe the transmitted signal spectrum, eye diagram, constellation diagram and transmitted symbols for a fixed SNR.
4. Observe the received signal spectrum, eye diagram, constellation and received symbols for a fixed SNR.
5. Repeat the step-4 for various values of the SNR and frequency offset.

Labview Description:

The path for the sub vi's of MPSK is described here:

Search where LabVIEW software is installed.

1. mod_psk transmitter.vi: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Modulation

2. MT Add AWGN.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Impairments

3. MT Apply IQ Impairments.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Impairments

4. MT Demodulate PSK.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation

5. MT measure PSK Quadrature Impairments.vi: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation

6. MT Calculate BER.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation

7. MT Calculate EbNo.vi: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation

8. Mod_align Tx and Rx bit sequence: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/support/examples

9. The case structure:

a. Transmitted Symbols/Demodulated Symbols:

MT bit stream to digital graph.vi: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General

b. TX Spectrum:

Mod_complex FFT(Mag_Phase).vi: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Analog/support

c. Constellation (Tx Output):

MT format Constellation.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General

d. Constellation (Rx Output):

mod_truncate filter transient.vi: Not available in Search

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/support/Examples

MT format Constellation.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General

e. Eye Diagram(Rx):

MT Format_Eye Diagram.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General

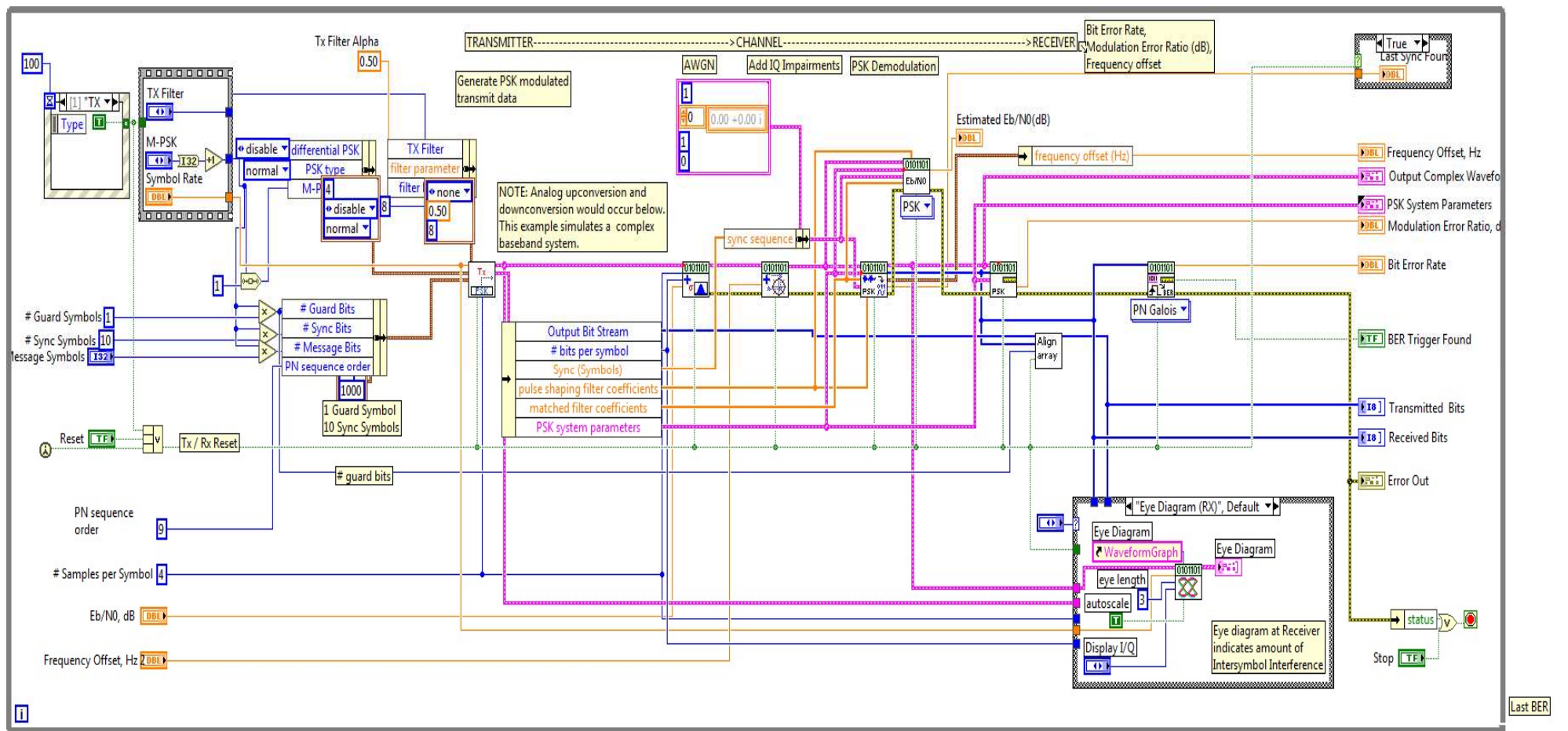


Fig1.1: Block diagram of MPSK Transceiver

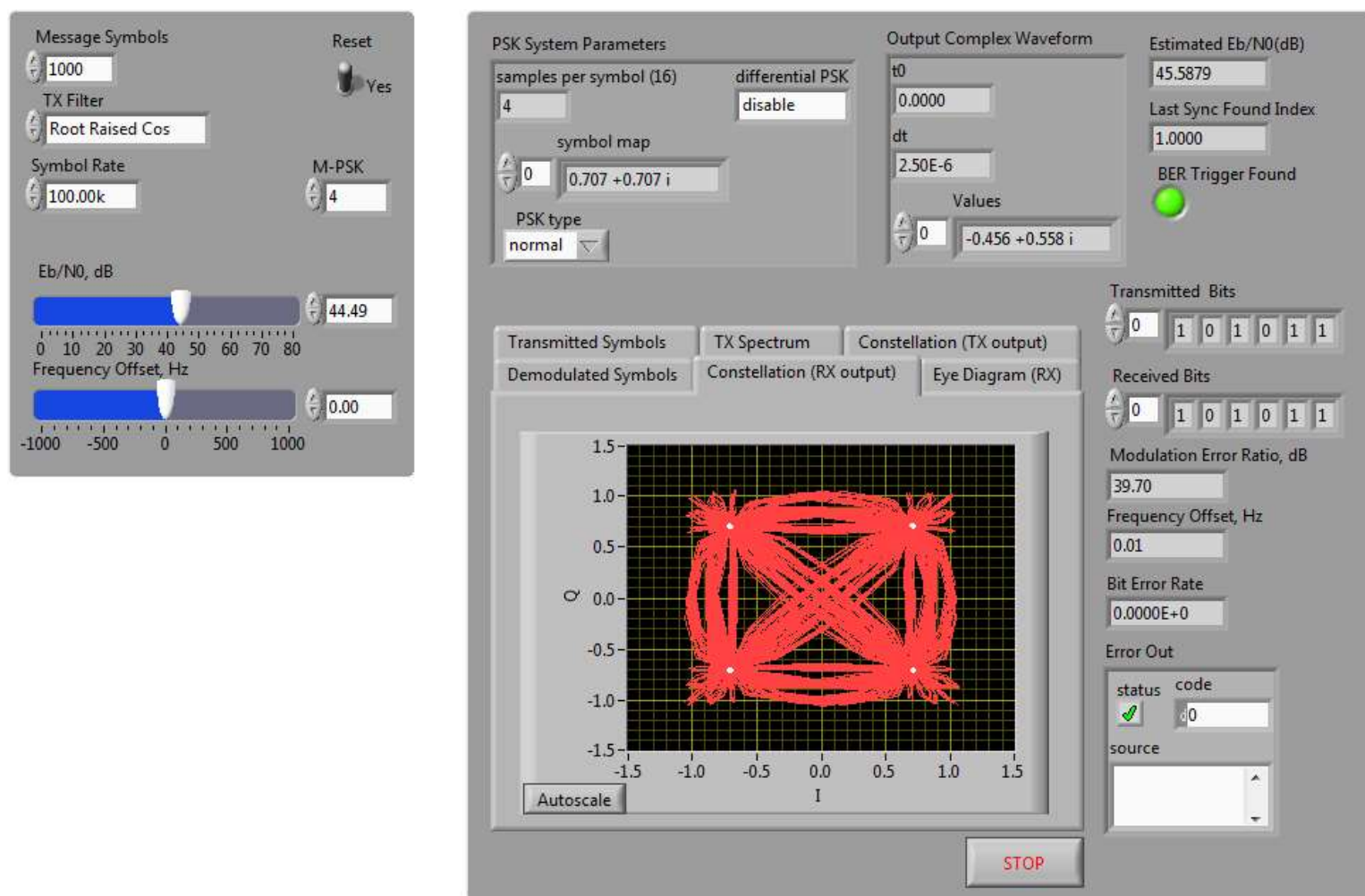


Fig1.2: Front panel for MPSK (M=4, QPSK)

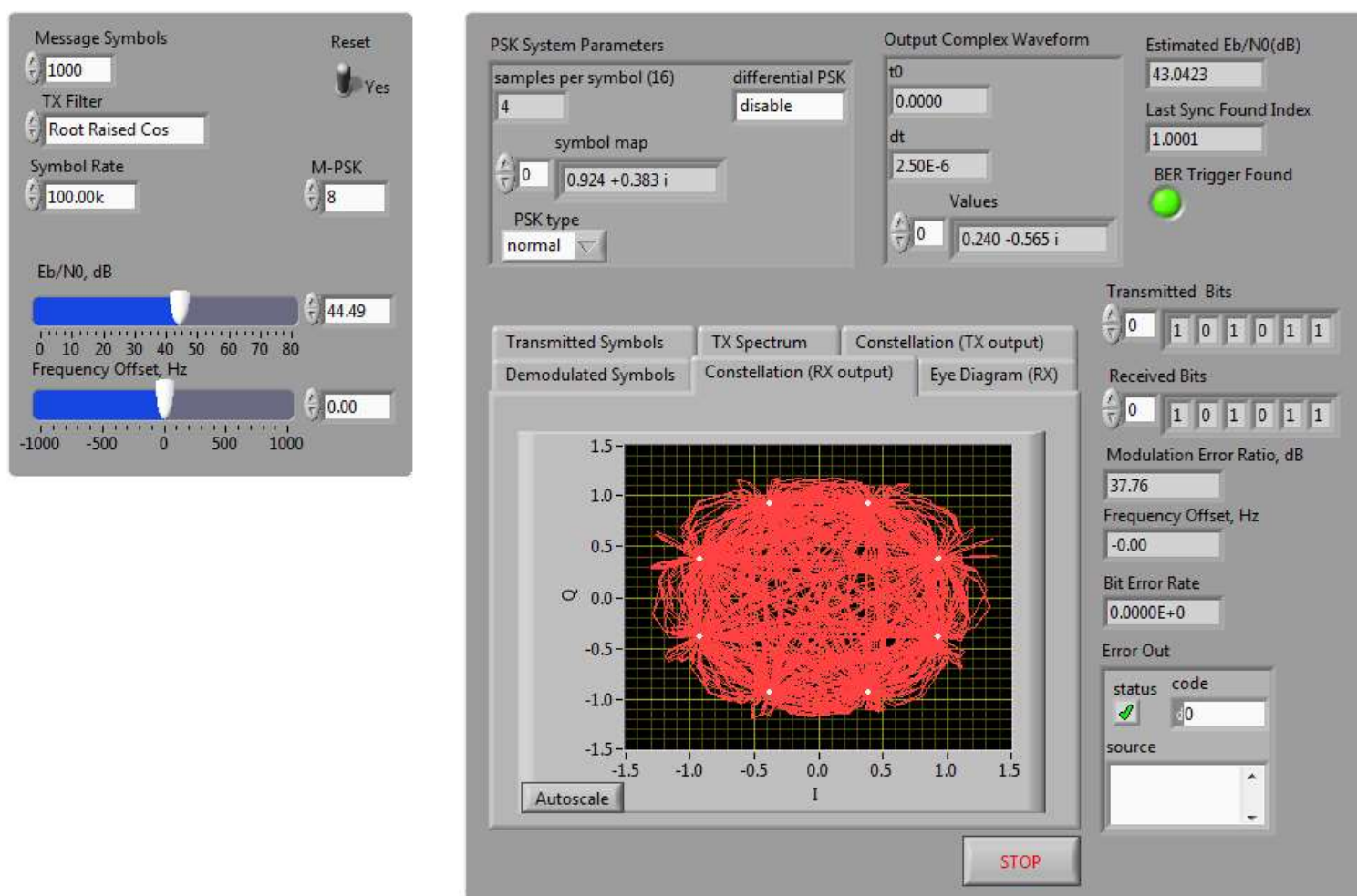


Fig1.3: Front panel for MPSK (M=8, 8-PSK)

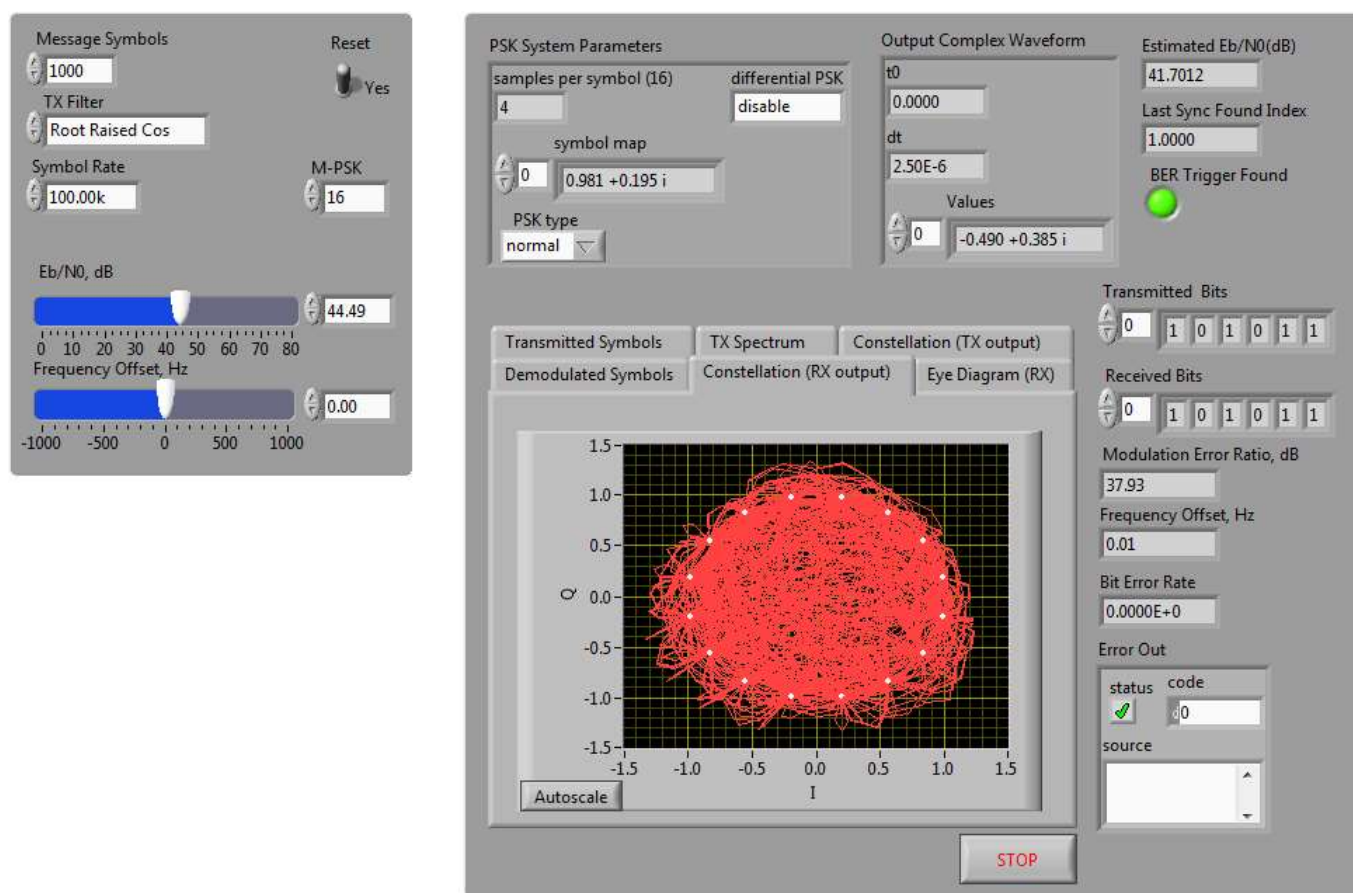


Fig1.4: Front panel for MPSK (M=16, 16-PSK)

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS				
Trial	M (No. of phases)	SNR (dB)	Frequency Offset (Hz)	Tx Spectrum (shape/BW)	Tx Constellation (pattern)	Rx Constellation (pattern)	Eye Diagram (Rx) (open/closed)	BER
1	4	10	0					
2	8	10	0					
3	16	10	0					
4	16	0	0					
5	16	20	0					
6	16	10	100					

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 2

M-QAM transceiver Design using Labview software

Aim: To implement M-QAM transceiver implementation using Labview software

Tools required: Desktop or Laptop with Labview software

Theory: Quadrature Amplitude Modulation

A variety of communication protocols implement quadrature amplitude modulation (QAM). Current protocols such as 802.11b wireless Ethernet (Wi-Fi) and digital video broadcast (DVB), for example, both utilize 64-QAM modulation. In addition, emerging wireless technologies such as Worldwide Interoperability for Microwave Access (WiMAX), 802.11n, and HSDPA/HSUPA (a new cellular data standard) will implement QAM as well. Thus, understanding QAM is important because of its widespread use in current and emerging technologies.

QAM involves sending digital information by periodically adjusting the phase and amplitude of a sinusoidal electromagnetic wave. Each combination of phase and amplitude is called a symbol and represents a digital bitstream. This tutorial first covers the hardware implementation required to constantly adjust the phase and amplitude of a carrier wave. The tutorial also discusses the binary value associated with each symbol.

Quadrature amplitude modulation (QAM) requires changing the phase and amplitude of a carrier sine wave. One of the easiest ways to implement QAM with hardware is to generate and mix two sine waves that are 90 degrees out of phase with one another. Adjusting only the amplitude of either signal can affect the phase and amplitude of the resulting mixed signal.

Procedure:

1. The basic block diagram of M-QAM transceiver in Labview is shown in figure 2.1.
2. The corresponding front panel in Labview is shown in figure 2.2, 2.3 and 2.4 for M=4,8 and 16 respectively.
3. Vary the M-levels in front panel (Ex M=2, 4,8,16 etc) and observe the transmitted signal spectrum, eye diagram, constellation diagram and transmitted symbols for a fixed SNR.
4. Observe the received signal spectrum, eye diagram, and constellation and received symbols for a fixed SNR.
5. Repeat the step-4 for various values of the SNR and frequency offset.

Labview Description:

The path for the sub vi's of MQAM is described here:

Search where LabVIEW software is installed.

1. **mod_QAM Transmitter.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Support/Examples
2. **MT Add AWGN.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Impairments

3. **MT Apply IQ Impairments.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Impairments
4. **MT Demodulate QAM.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation
5. **MT measure QAM Quadrature Impairments.vi: Not available in Search**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation
6. **MT Calculate BER.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation
7. **MT Calculate EbN0.vi: Not available in Search**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation
8. **MT Modulate QAM: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Modulation
9. **MT Measure Rho: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/Demodulation
10. **mod_align TX and RX bit sequence: Not available in Search**
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/support/examples
11. **The case structure:**
 - a. Transmitted Symbols/Demodulated Symbols:
MT Bit Stream to Digital Graph.vi: Not available in Search
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General
 - b. Constellation (Tx Output):
MT Format Constellation.vi: Available in search or Quick drop (ctrl + Spacebar)
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General
 - c. Constellation (Rx Output):
mod_Truncate Filter Transient.vi: Not available in Search
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/support/Examples
MT Format Constellation.vi: Available in search or Quick drop (ctrl + Spacebar)
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General
 - d. Eye Diagram (Rx):
MT Format_Eye Diagram.vi: Available in search or Quick drop (ctrl + Spacebar)
Path: National Instruments/Labview2020/vi.lib/addons/Modulation/Digital/General

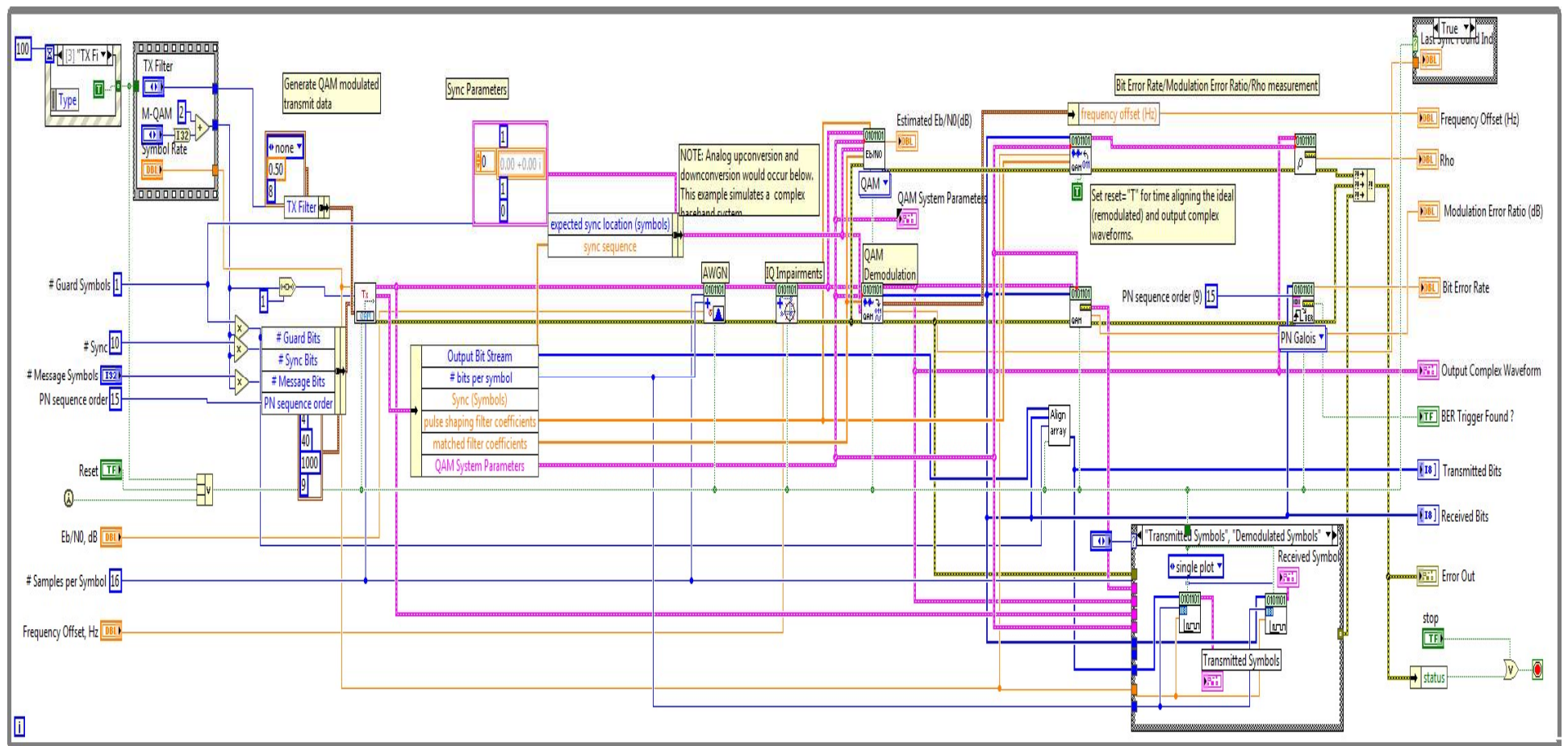


Fig2.1: Block diagram of MQAM Transceiver

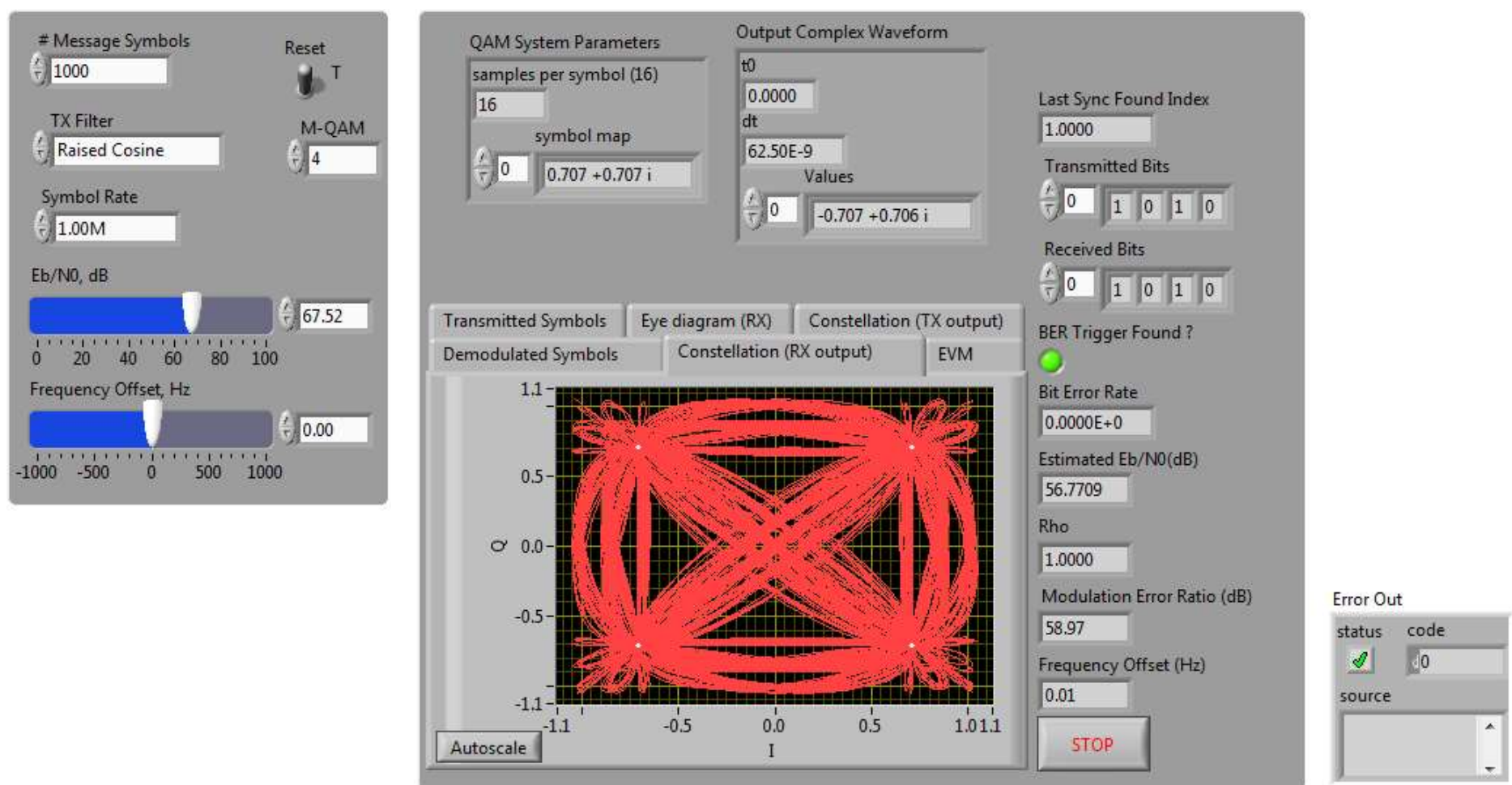


Fig2.2: Front panel for MQAM (M=4, 4-QAM)

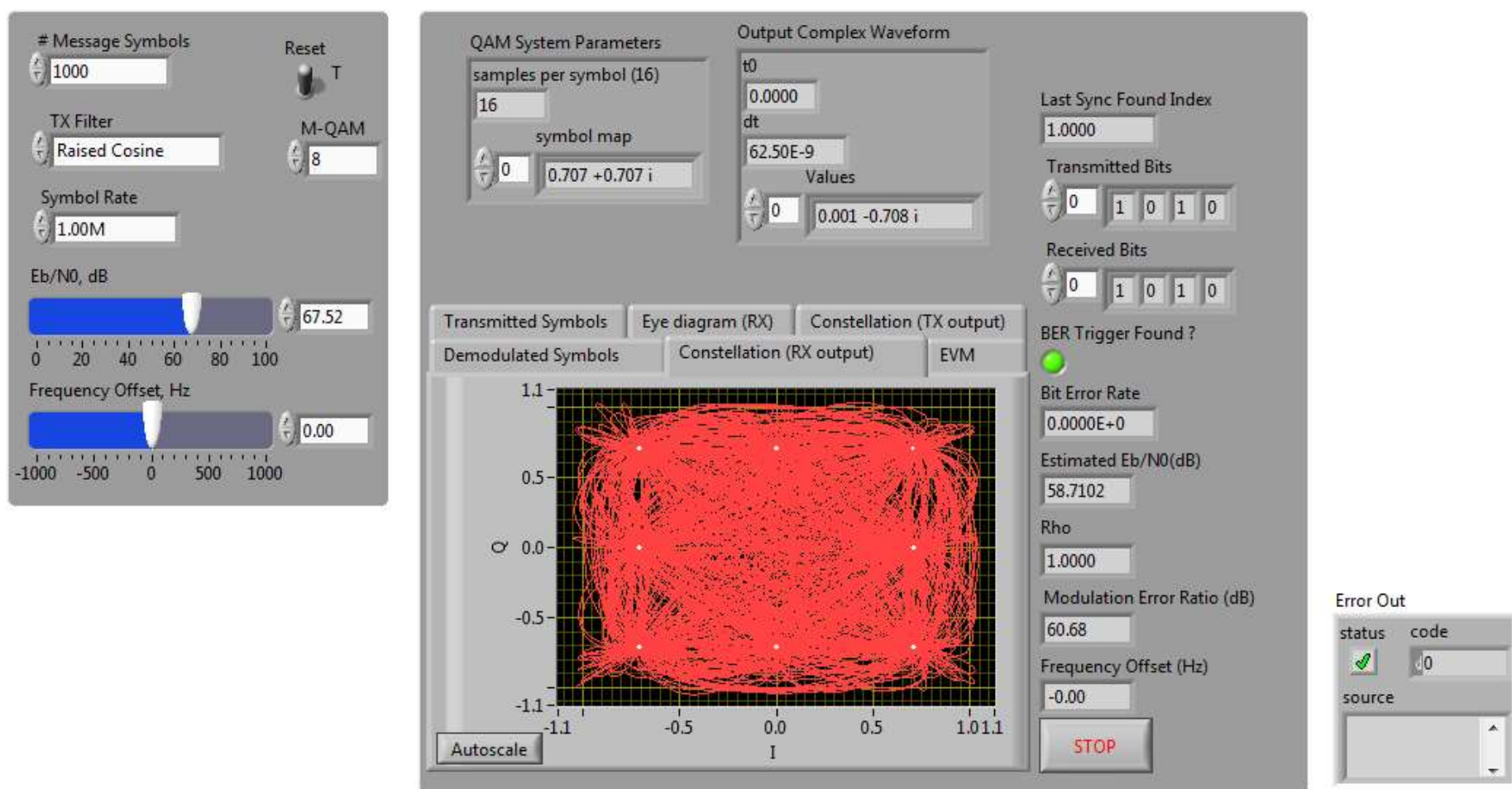


Fig2.3: Front panel for MQAM (M=8, 8-QAM)

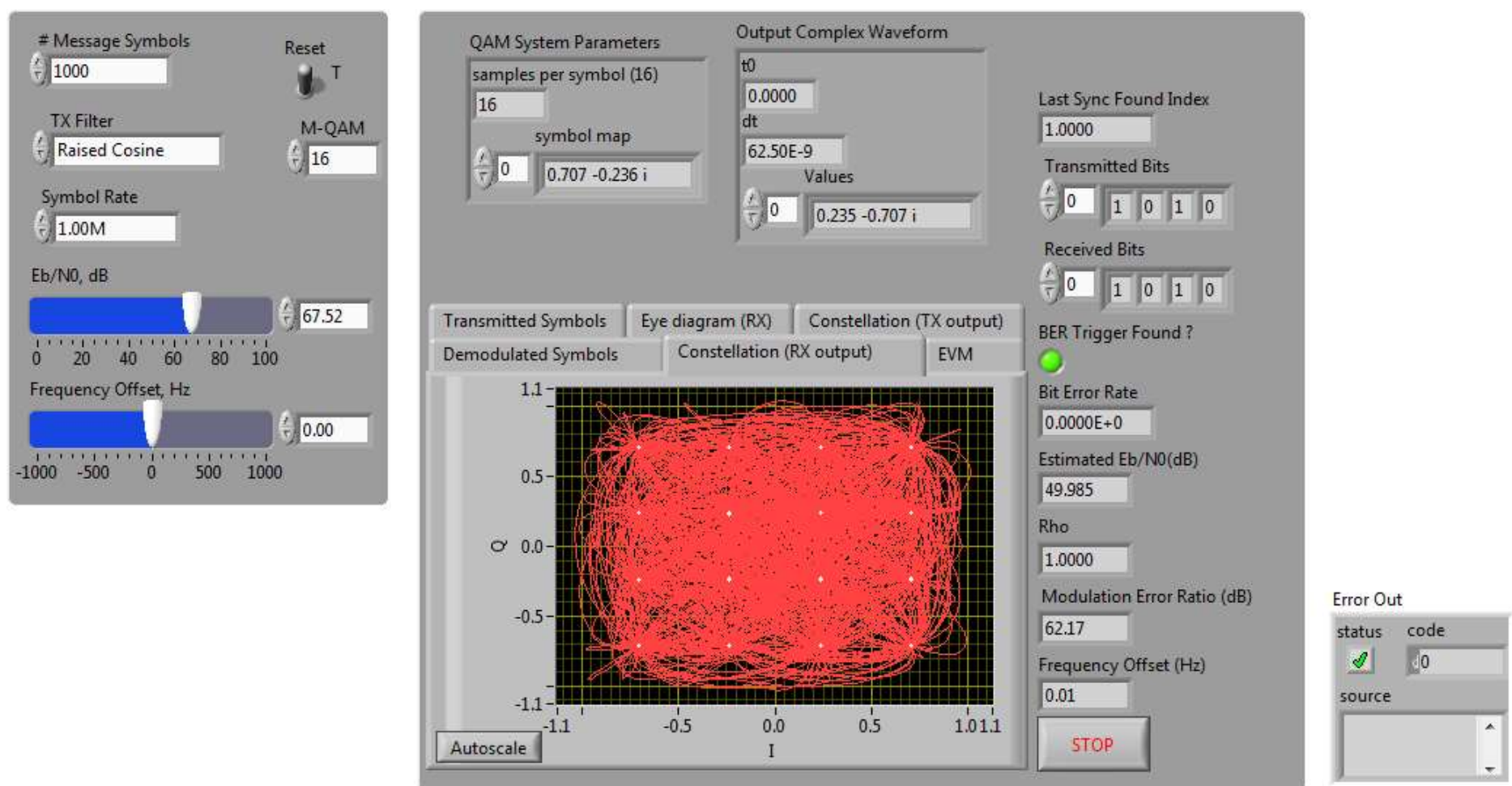


Fig2.4: Front panel for MQAM (M=16, 16-QAM)

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS				
Trial	M (QAM order)	SNR (dB)	Frequency Offset (Hz)	Tx Spectrum (shape/BW)	Tx Constellation (pattern)	Rx Constellation (pattern)	Eye Diagram (Rx) (open/closed)	BER
1	4	10	0					
2	8	10	0					
3	16	10	0					
4	16	0	0					
5	16	20	0					
6	16	10	100					

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 3

ISI with PAM modulation technique using MATLAB

Aim: To simulate ISI effect with PAM modulation technique using MATLAB

Tools required: Desktop or Laptop with MATLAB software

Procedure:

1. Generate PAM symbols and modulate
2. Upsample the modulated symbols and pass it through SRRC pulse shaping
3. Add AWGN noise for given SNR value
4. Matched filter with SRRC pulse shape
5. Symbol rate sampler and demodulation
6. Generating eye plot at the matched filter output

Theory:

The basic pulse amplitude modulation (PAM) system is enhanced in Figure 3.1 by adding an upsampler ($\uparrow L$), pulse shaping function ($p[n]$) at the transmitter and a matched filter ($g[n]$), down-sampler ($\downarrow L$) combination at the receiver. In this model, a random stream of source bits is first segmented into k -bit wide symbols that can take any value from the set $m \in \{1, 2, \dots, M\}$. The simulation code directly starts by generating a random set of symbols, that goes into the modulation mapper. Pulse amplitude modulation (MPAM) mapping and de-mapping, are considered here for simulation. An MPAM modulator maps the k -bit information symbols to one of the $M = 2^k$ distinct signaling levels. The MPAM modulated symbols are shown in Figure 3.2. The sample waveforms at each stage of figure 3.1 are shown for reference in Fig3.3 onwards.

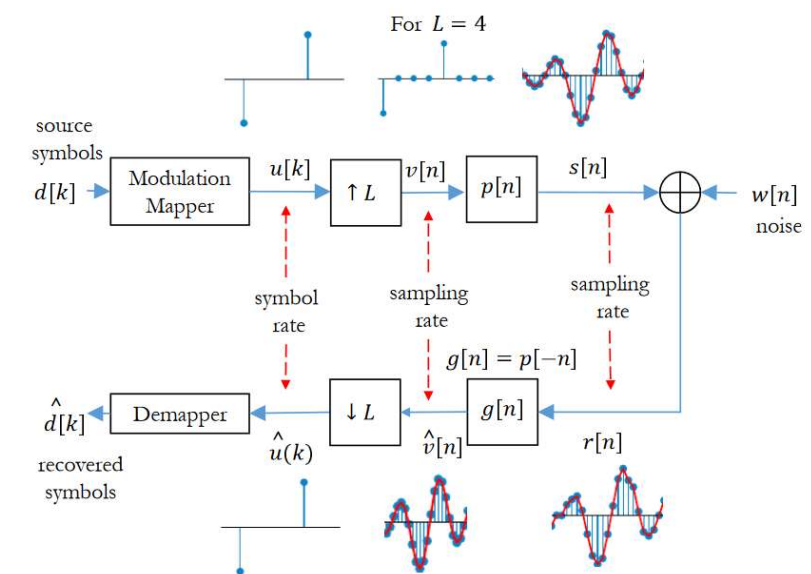


Fig. 3.1: DSP implementation of a PAM modulation system with pulse shaping and matched filtering

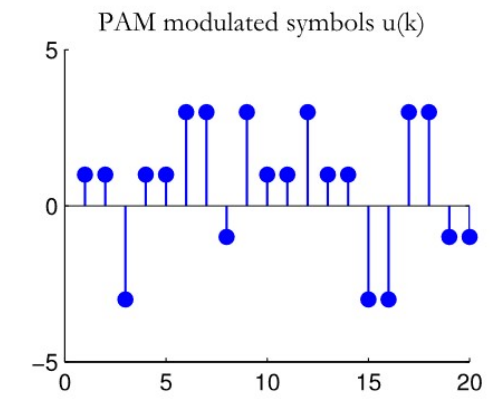


Fig3.2: MPAM modulated symbols

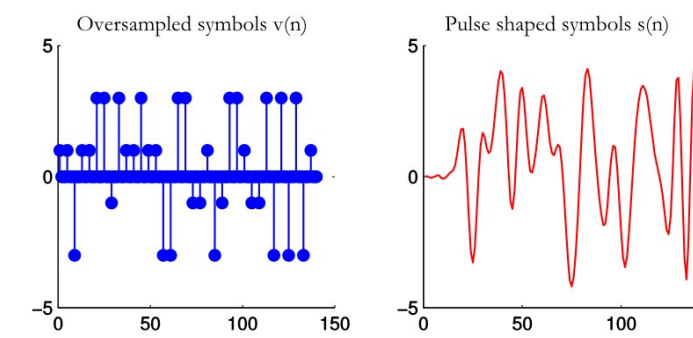


Fig3.3: Modulated symbols upsampled by 4 (left) and the SRRC pulse shaping filter output (right)

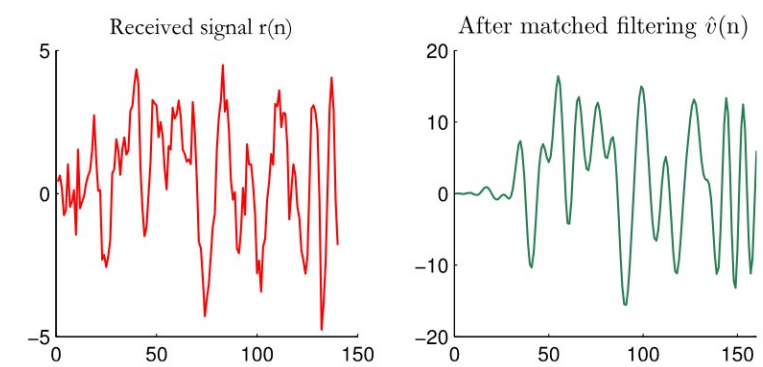


Fig. 3.4: Received signal with AWGN noise (left) and the output of the matched filter (right)

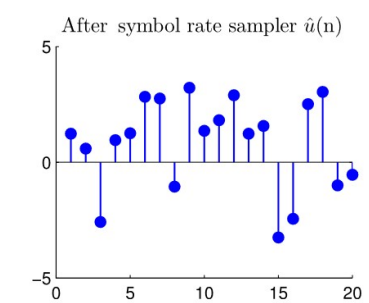


Fig3.5: Down sampling - output of symbol rate sampler

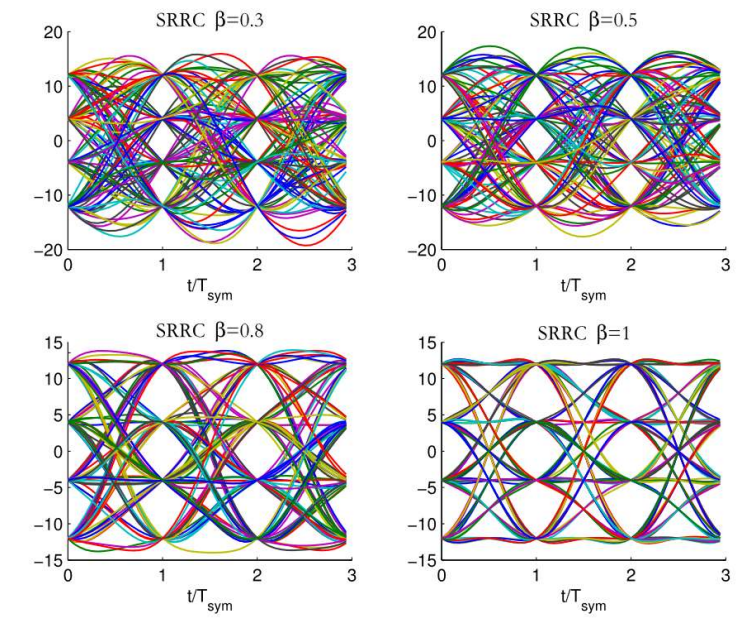


Fig. 3.6: Eye plot at the matched filter output just before the symbol sampler at the receive

Conclusion:

INPUT PARAMETERS					OUTPUT PARAMETERS		
Trial	M (PAM order)	Roll-off factor (α)	Upsampling factor (L)	SNR (dB)	Eye Opening (wide/narrow/closed)	ISI Observed? (Y/N)	BER
1	2	0.2	4	20			
2	2	0.5	4	20			
3	2	1.0	4	20			
4	4	0.5	4	20			
5	4	0.5	4	5			
6	4	0.5	4	0			

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 4

Equalization with QAM Modulation using Labview

Aim: Demonstrate the implementation of Equalization in an QAM Modulation-Demodulation System

Tools required: Desktop or Laptop with Labview software

Theory:

Equalization in QAM modulation is used to combat channel impairments like inter-symbol interference (ISI) and multipath fading, which distort the signal during transmission. It restores the signal's integrity, ensuring correct data detection. Channel Effects (noise, fading, multipath) distort the QAM signal, causing symbol detection errors. Equalization mitigates these effects by compensating for channel-induced distortions. Common techniques used are:

- Linear Equalizers: Zero Forcing (ZF) and Minimum Mean Square Error (MMSE).
- Decision Feedback Equalizers (DFE): Uses previous symbols to remove ISI.
- Adaptive Equalizers: Dynamically adjusts to changing conditions.
- MLSE: Optimal but computationally heavy.

Equalization is Critical for high-order QAM (64-QAM, 256-QAM) in wireless and high-speed communication systems like 5G, Wi-Fi, and DSL. Equalization ensures reliable data transmission in QAM systems by counteracting channel distortions.

Procedure:

1. The front panel in Labview is shown in figure 4.1b, for M=16.
2. The basic block diagram of M-QAM with equalization in Labview is shown in figure 4.2.
3. Vary the M-levels in front panel (Ex M=4,8,16,256 etc) and observe the Equalizer Taps, LMS Error, constellation diagram before equalization and constellation diagram after equalization.
4. Observe the received data, BER and modulation error ratio.
5. Repeat the step-4 for various values of the SNR and equalizer length.

LabVIEW Description:

This program applies adaptive feedforward software equalization using the least-mean-squared (LMS) algorithm to the QAM-demodulated input complex waveform. Use the Specific Coefficients instance of this polymorphic VI if the equalizer coefficients are already known or when the initial conditions for the equalizer coefficients must be specified. Use the Specify Length polymorphic instance when only the length of the QAM feedforward equalizer (in symbols) must be specified.

The following flow chart (Fig 4.1a) illustrates demodulator-to-equalizer data flow.



Fig4.1 a: Data flow illustration

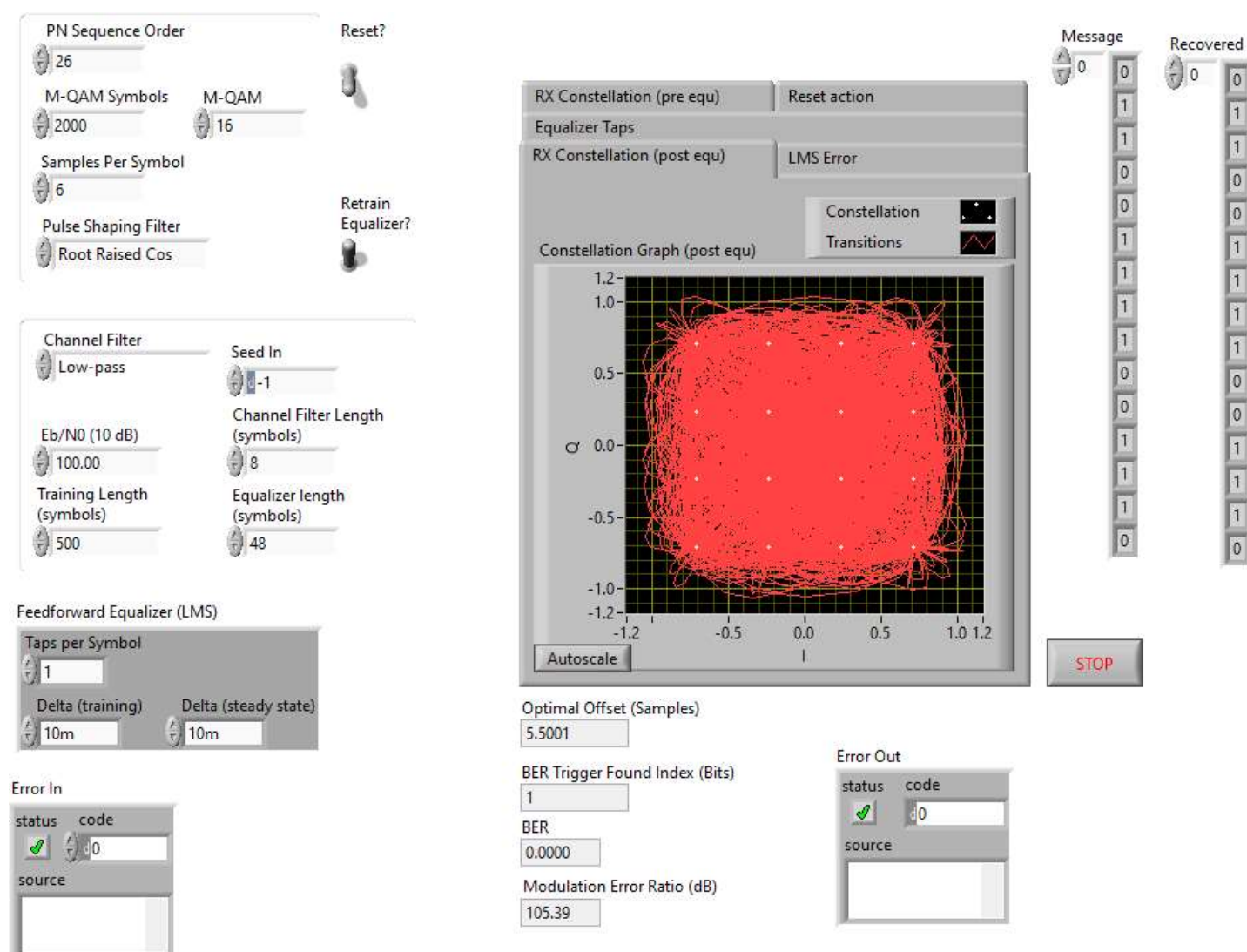


Fig4.1b: Front panel for MQAM with Equalizer

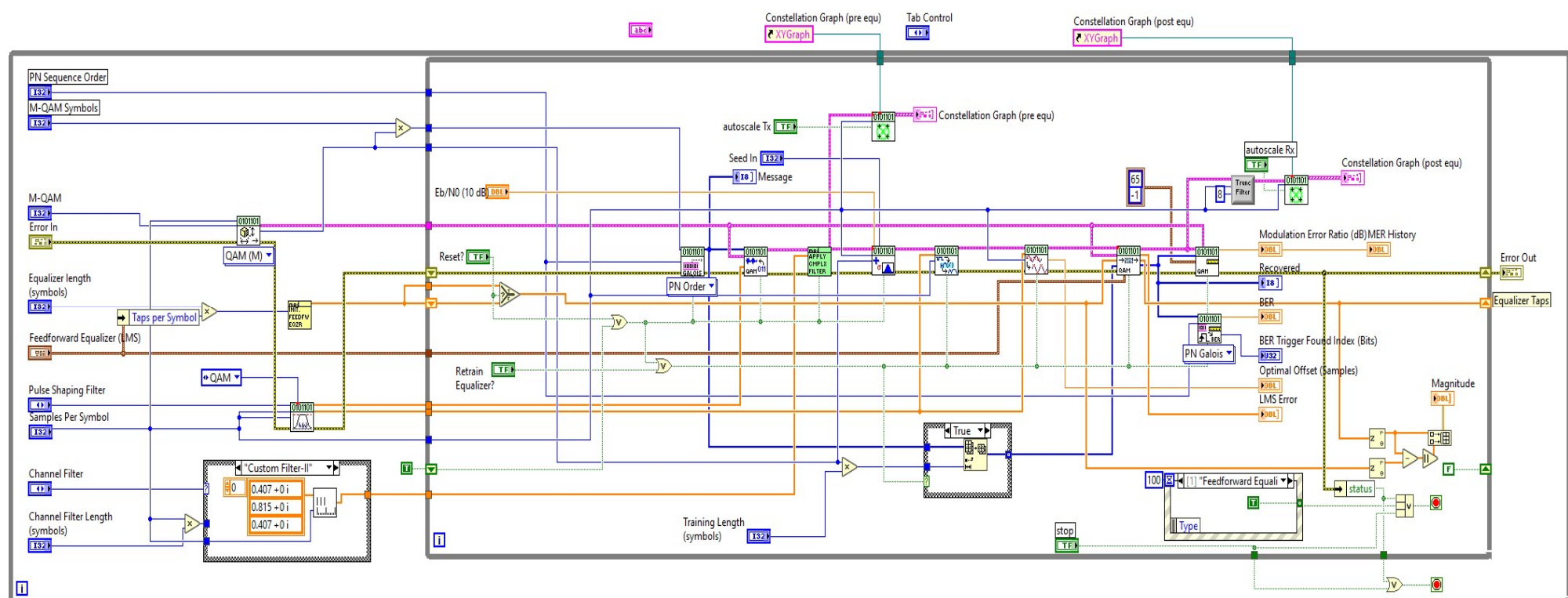


Fig4.2: Block diagram for MQAM with Equalizer

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS				
Trial	M (QAM order)	SNR (dB)	Equalizer Length (taps)	Equalizer Tap Values	LMS Error (converged value)	Constellation (before/after Eq.)	BER	Modulation Error Ratio (MER)
1	16	15	5					
2	64	15	5					
3	256	15	5					
4	16	5	5					
5	16	25	5					
6	16	15	11					

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 5

Transmitter and Receiver Implementation of Direct Sequence Spread Spectrum and Frequency Hopped Spread Spectrum

Aim: To implement Transmitter and Receiver for Direct Sequence Spread Spectrum and Frequency Hopped Spread Spectrum

Tools required: Desktop or Laptop with Matlab software

Theory:

Spread-spectrum (SS) technology uses a sequential noise-like signal to spread the normally narrowband information signal over a relatively wide band of radio frequencies. The receiver correlates the received signals to retrieve the original information signal. Using wideband, noise-like signals to “spread the spectrum” used makes the transmissions hard to detect, intercept, jam, or demodulate. This is why the military has used spread-spectrum methods for many years.

The Direct Sequence Spread Spectrum (DSSS) Technique

In the Direct Sequence Spread Spectrum (DSSS) technique, the signal being transmitted is divided and injected with multiple frequencies within a particular frequency band as shown in figure 5.1. The original data is mixed with redundant data bits or code, called chips or chipping code, and the ratio of the chips to information is called the spreading ratio. A high spreading ratio indicates a wider bandwidth. As per Shannon’s theorem, the higher the bandwidth, the more immune the signal transmission is to crosstalk, interference, and noise.

The DSSS technique helps maintain secure signal transmission with a high signal-to-noise ratio (SNR) at the receiving end. And, the DSSS technique helps recover the original data even when a part of the transmitted data is corrupted. The DSSS method of modulation has a higher rate of transmission than the FHSS method. In terms of the signal transmission rate, DSSS is superior. However, DSSS is vulnerable to electromagnetic interference and noise produced by devices operating at the same frequency band. Output for DSSS at transmitter and receiver are shown in figures 5.2.

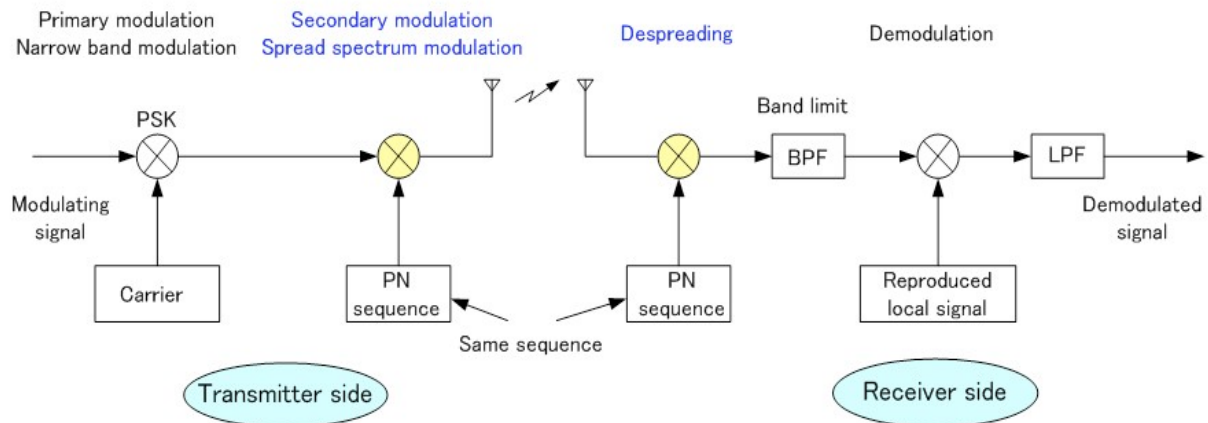


Fig 5.1- Transmitter and Receiver Implementation of Direct Sequence Spread Spectrum

The Frequency Hopping Spread Spectrum (FHSS) Technique

The spread spectrum technique in which the carrier frequency is changed according to the pseudo-random noise or sequence injected is called the frequency hopping spread spectrum (FHSS) technique as shown in figure 5.3. The pseudo-random sequence is only available with the transmitter and receiver. The signal transmitted is injected with noise, increasing the bandwidth of the original information. At the receiving end, the pseudo-random sequence is removed to retrieve the original information. The FHSS technique prevents the loss of data and limits noise, crosstalk, and electromagnetic interference, preserving the signal integrity and reliability of communications.

According to the frequency at which the RF frequency varies in signal communications, the FHSS technique is classified into either slow frequency hopping or fast frequency hopping. In fast frequency hopping, the hopping is carried out at a faster rate than the information bit rate. When the frequency hopping rate is less than the message bit rate, the spread spectrum technique is called slow frequency hopping. The frequency hopping spread spectrum technique is often implemented in wireless local area networks. Output for FHSS at transmitter and receiver are shown in figures 5.4.

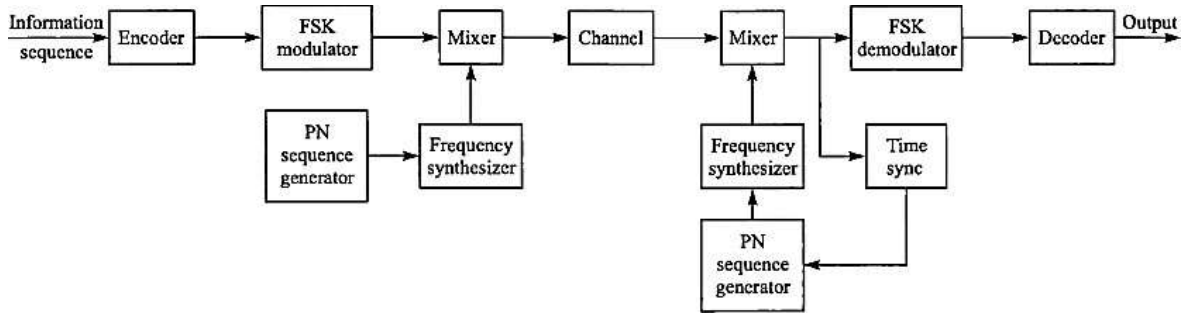
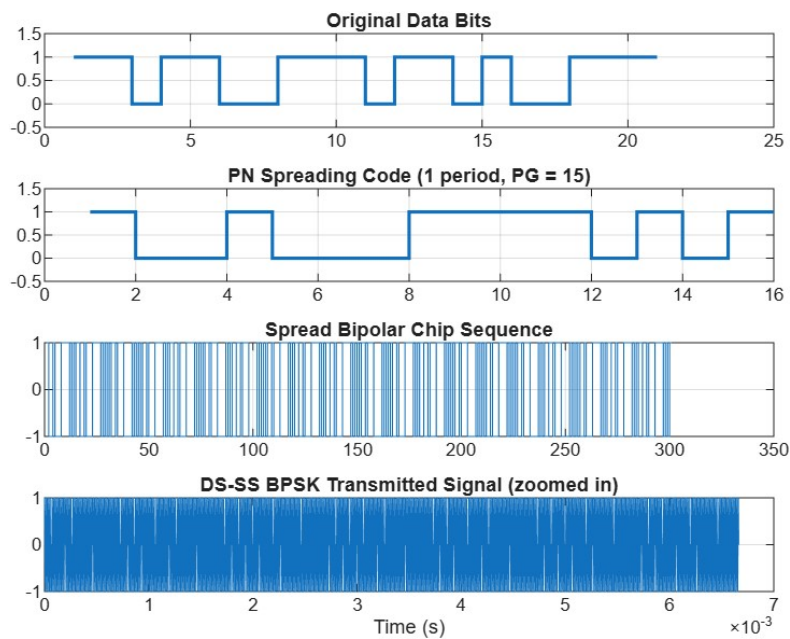


Figure 5.2: Transmitter and Receiver Implementation of Frequency Hopped Spread Spectrum

Output:

1. Direct Sequence Spread Spectrum (DSSS) Technique



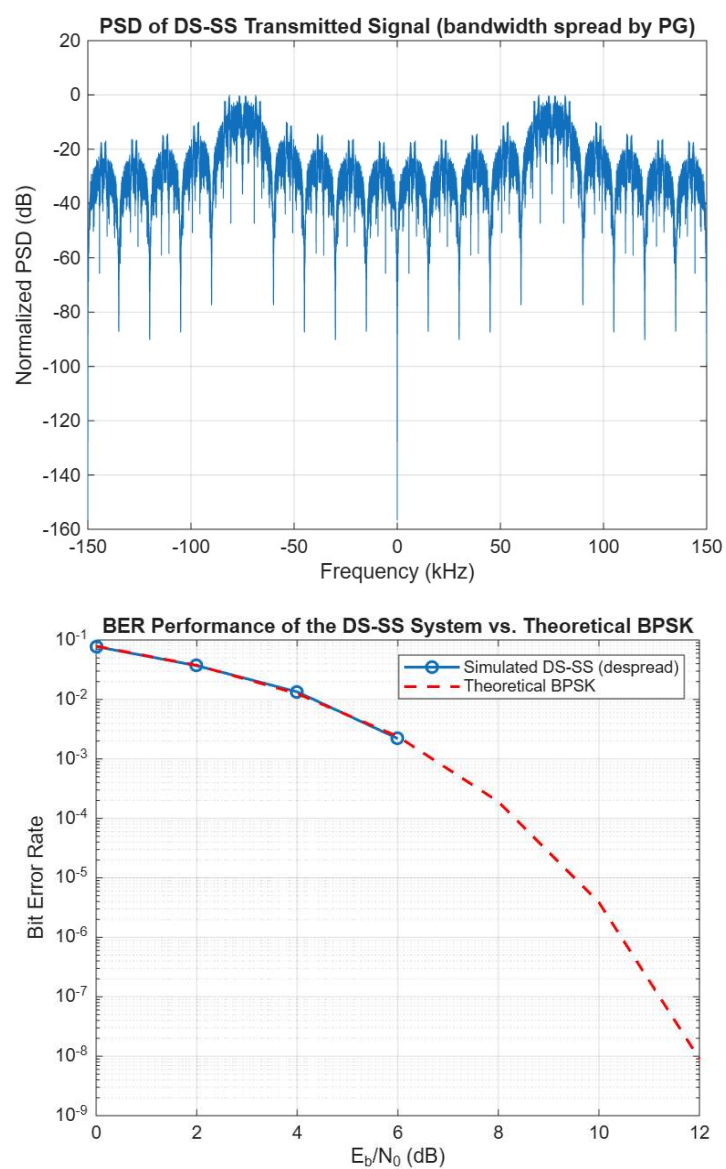
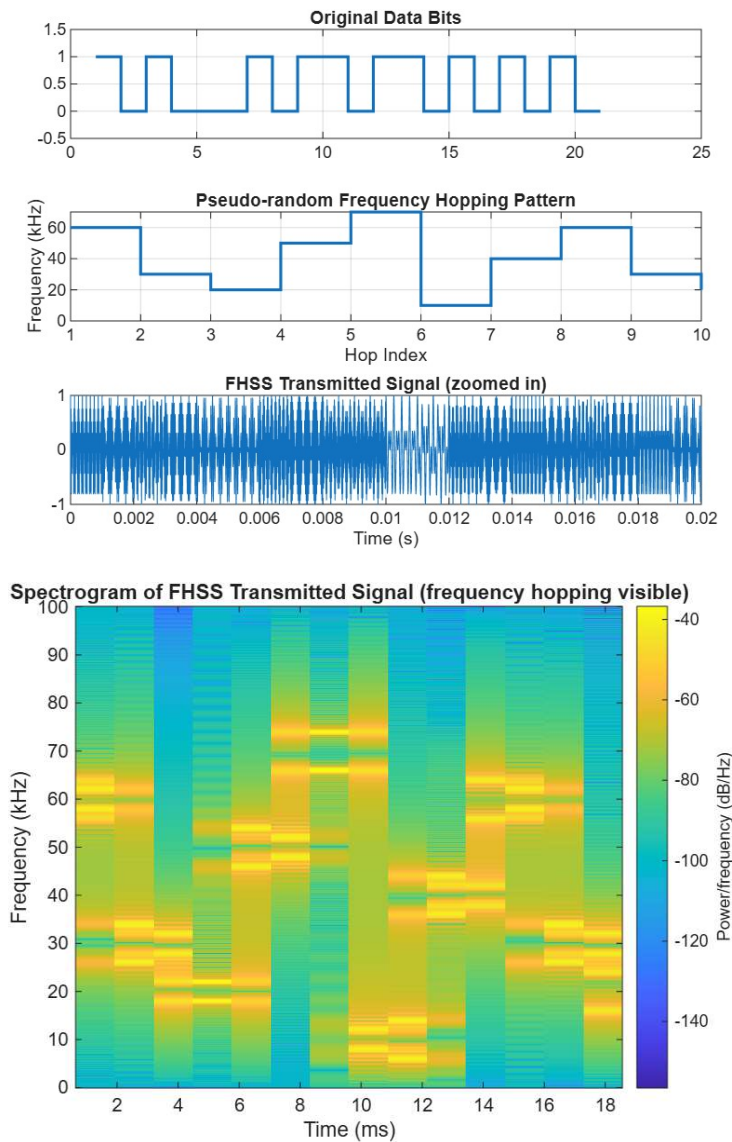


Figure 5.3: Direct sequence spread spectrum

2. Frequency Hopping Spread Spectrum (FHSS) Technique



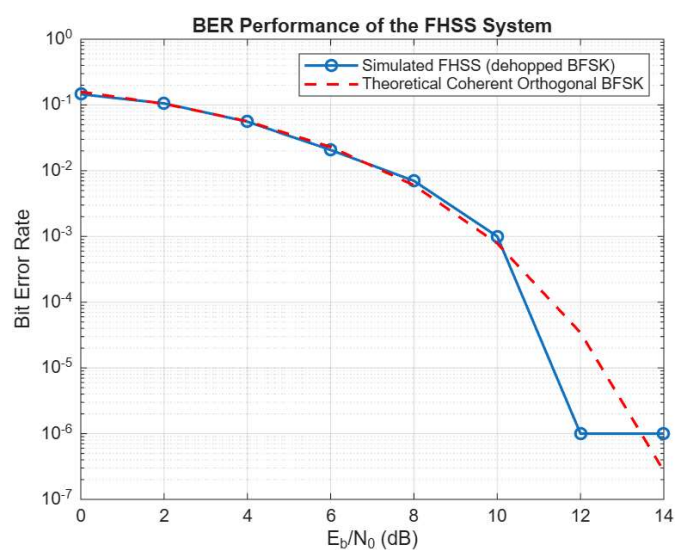


Figure 5.4: Frequency spread spectrum signals

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS			
Trial	Technique (DSSS/FHSS)	Spreading Code Length / No. of Hop Frequencies	Chip Rate / Hop Rate	Spread Signal (time domain)	Processing Gain (dB)	Occupied Bandwidth	PSD Observation
1	DSSS	7	Low				
2	DSSS	15	Low				
3	DSSS	31	High				
4	FHSS	4 hops	Slow hopping				
5	FHSS	8 hops	Slow hopping				
6	FHSS	8 hops	Fast hopping				

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Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 6

Symbol Timing Estimation using MATLAB

Aim: To demonstrate the symbol timing estimation

Tools required: Desktop or Laptop with Matlab software

Theory:

Symbol timing synchronization works by selecting the best sample from each group of samples to represent the transmitted symbol. The goal is to produce L samples per symbol at the matched filter output, with one of those samples aligned with the maximum eye opening. The block diagram of Symbol timing synchronization consists of Timing Error Detector, Loop Filter, Interpolator and Controller (Fig 6.1).

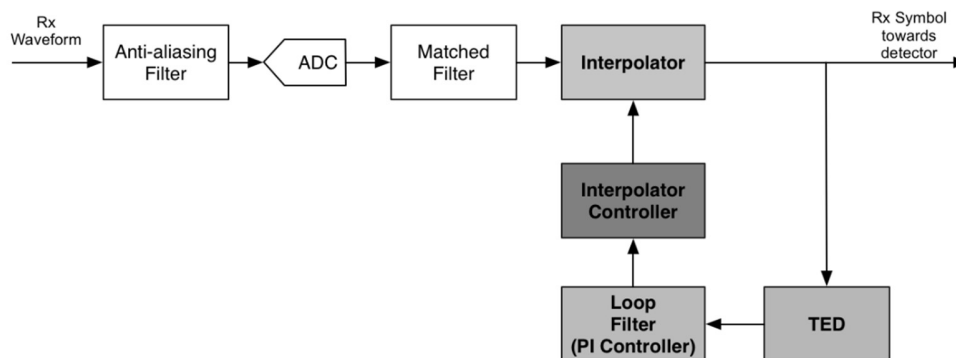
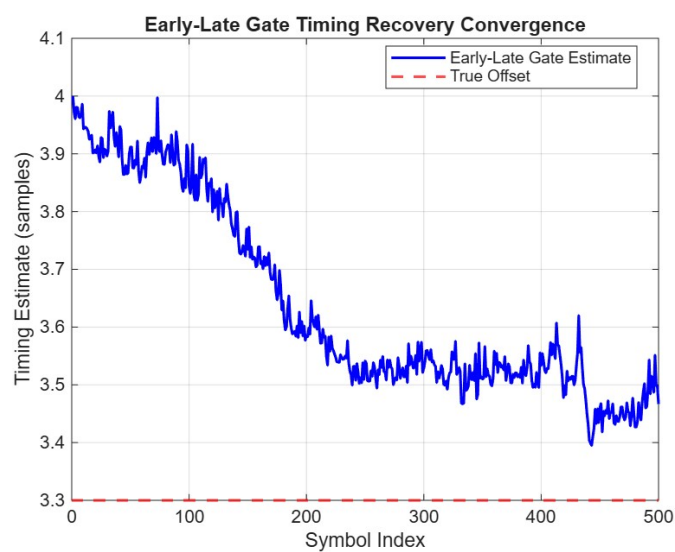
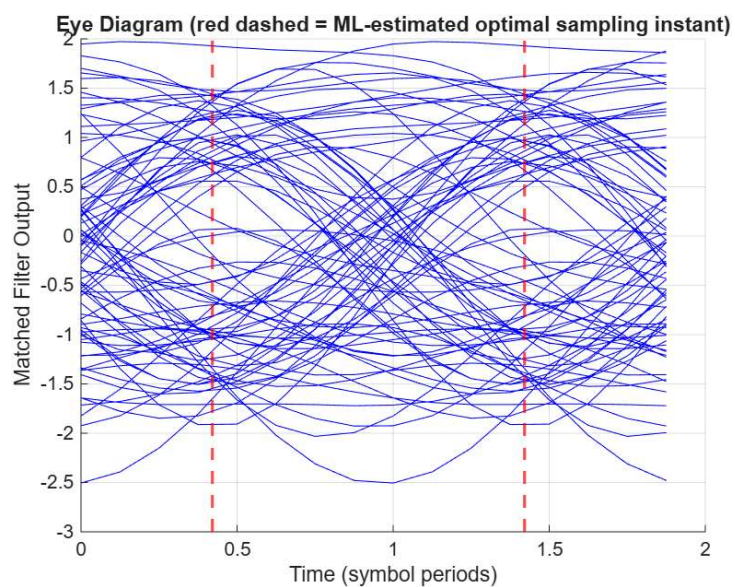


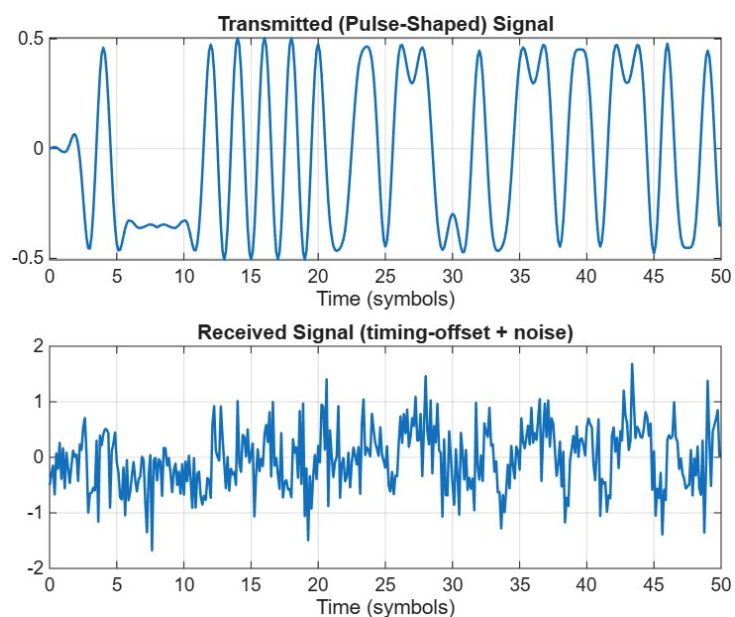
Fig6.1: Block diagram of symbol timing estimation

Algorithm:

1. Design a Square-root Raised Cosine filter for pulse shaping
2. Generate the transmission of a 2-PAM/4-PAM symbol sequence
3. Add a random channel propagation delay in units of sampling intervals
4. Develop a receiver that does not perform symbol timing synchronization. (The matched filter block precedes the symbol timing recovery loop).
5. Add an arbitrary timing used by the receiver to select samples from the incoming sequence and pass them to the decision module
6. Plot the scatter plots (expected results are shown in fig 6.2a, and b)
7. Design symbol timing offset exactly and plot the received symbols and scatter plots.

Expected plots





Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS		
Trial	Modulation (2-PAM/4-PAM)	Timing Offset (fraction of T_s)	SNR (dB)	Scatter Plot (No Sync)	Scatter Plot (With Sync)	Symbol Error Observed
1	2-PAM	0	20			
2	2-PAM	0.2	20			
3	2-PAM	0.5	20			
4	4-PAM	0.2	20			
5	4-PAM	0.2	10			
6	4-PAM	0.5	10			

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Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 7

Frequency selective and non-selective fading channel simulation using MATLAB

Aim: To observe the impact on the received signals after passing through frequency selective fading and frequency non-selective fading channel.

Tools required: Desktop with Matlab

Theory:

Frequency nonselective fading channel: If all frequency components of a transmitted signal that fall within the coherence bandwidth B_{cb} will fade simultaneously. If the transmitted signal has a bandwidth $W < B_{cb}$, the channel is called frequency

nonselective. Thus, at any instant in time, all frequency components of the transmitted signal fade simultaneously. On the other hand, if the transmitted signal has a bandwidth $W > B_{cb}$, the frequencies in the signal separated by an amount greater than B_{cb} will be affected differently by the channel. Hence, at any instant, some frequency components in the transmitted signal may fade, whereas other frequency components may not. In such a case, the channel is said to be frequency selective.

$$B_{cb} = \frac{1}{T_m}$$

Where, channel coherence bandwidth (B_{cb}) and multipath (time) spread of the channel (T_m)

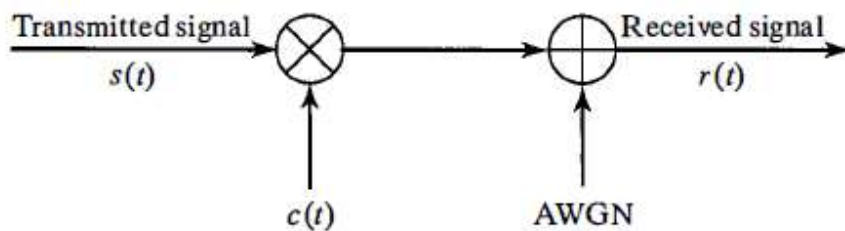


Fig-7.1 Model of a frequency nonselective time-varying channel with AWGN.

The frequency nonselective, slowly fading channel model as shown in fig 7.1 applies to many physical radio channels used for digital communications when the bandwidth W of the transmitted signal satisfies the condition $W \ll B_{cb}$. However, there are communication systems in which the

transmitted signal bandwidth $W \gg B_{cb}$ so that the channel is frequency selective. In such a case, a more complex channel model must be employed as shown in fig 7.2.

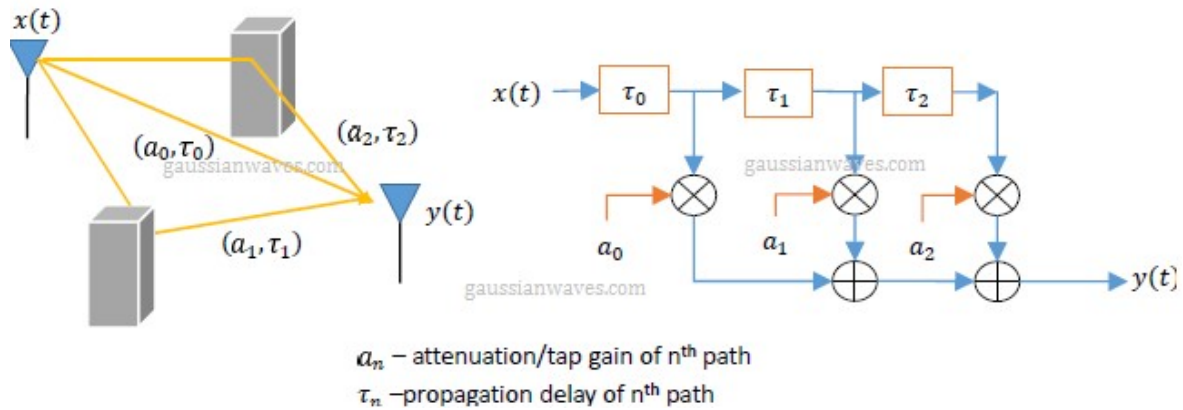
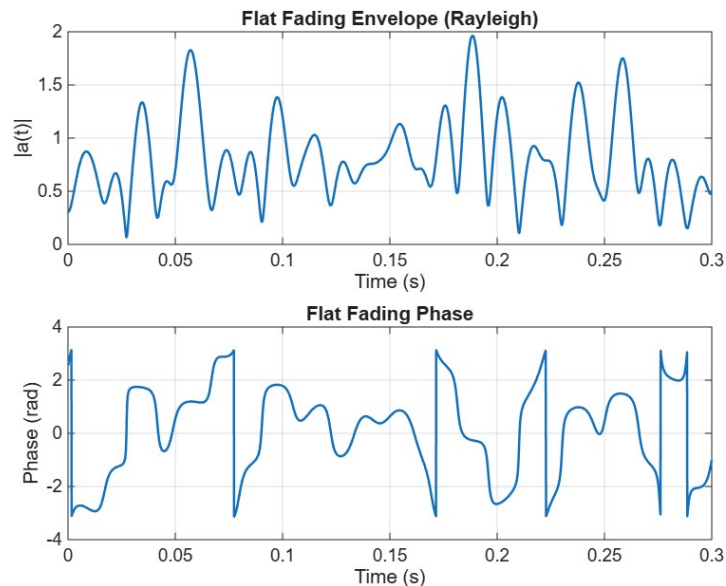
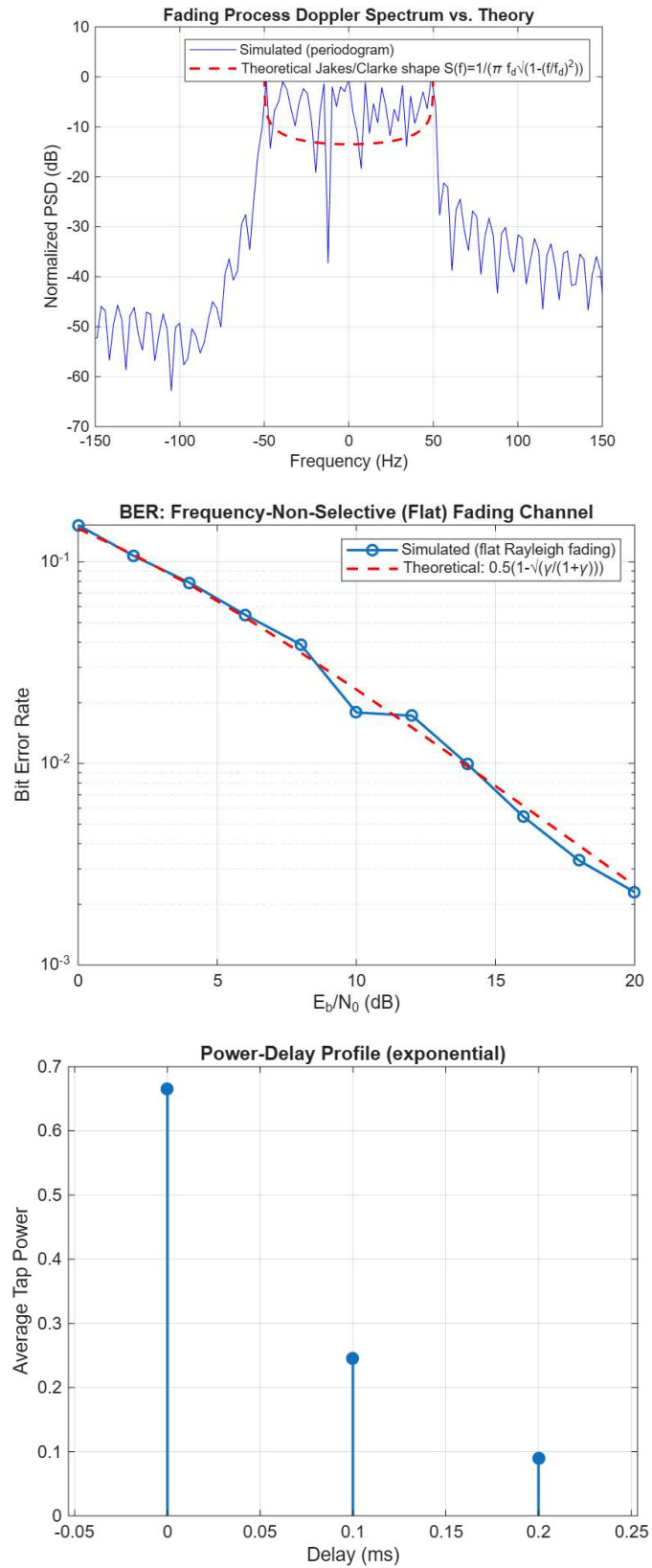


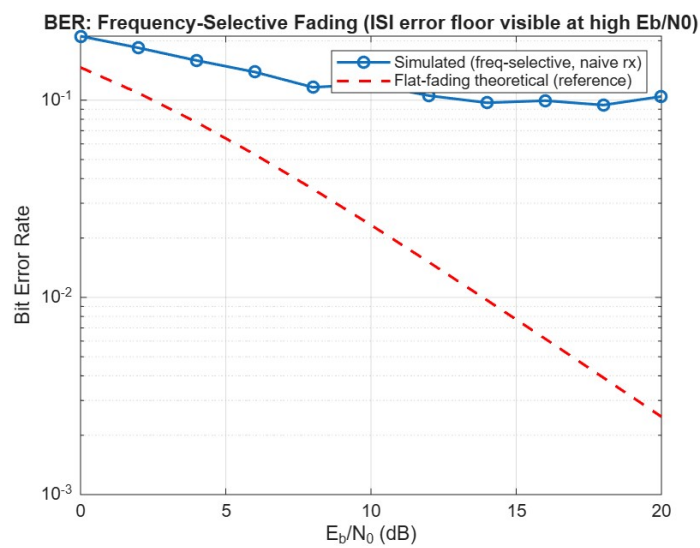
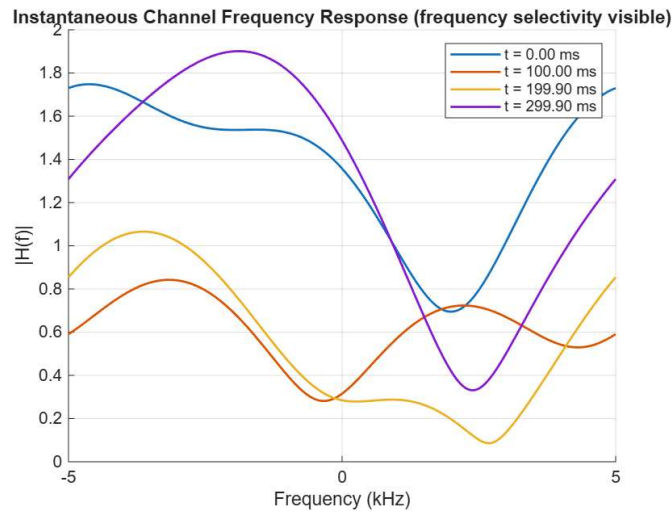
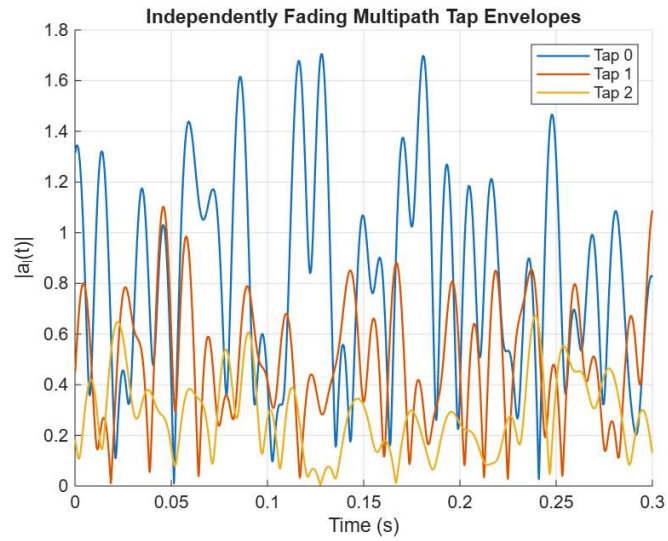
Fig7.2- Model of a frequency nonselective time-varying channel with AWGN.

A **frequency selective fading channel** is a type of fading in wireless communications where different frequencies of the transmitted signal experience varying amounts of attenuation. This happens cause the signal's different frequency components take multiple paths to the receiver (multipath propagation), causing some to interfere constructively while others interfere destructively.

Expected output







Conclusion:

INPUT PARAMETERS					OUTPUT PARAMETERS		
Trial	Fading Type	No. of Paths / Coherence BW	Path Delays / Gains (or Doppler shift)	SNR (dB)	Frequency Response / Signal Amplitude	Received Signal Waveform	BER
1	Frequency-selective	3 paths	[0, 1, 2] samples	15			
2	Frequency-selective	5 paths	[0,1,2,3,4] samples	15			
3	Frequency-selective	3 paths	[0, 1, 2] samples	5			
4	Frequency-non-selective	Bcb >> signal BW	Doppler = 10 Hz	15			
5	Frequency-non-selective	Bcb >> signal BW	Doppler = 50 Hz	15			
6	Frequency-non-selective	Bcb >> signal BW	Doppler = 50 Hz	5			

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 8

Carrier Phase Estimation: Decision-Directed and Non-Decision-Directed Loops

Aim: Carrier Phase Estimation - ML Carrier Phase Estimation, The PLL, Effect of Additive Noise on the Phase Estimate, Decision-Directed and Non-Decision-Directed Loops.

Tools required: Desktop with Matlab

Theory: The various stages of this experiment is described as follows.

1. QPSK Symbol Generation

QPSK Modulation:

The transmitted QPSK symbols can be represented mathematically as:

$$s[n] = A \cdot e^{j(\theta[n]+\phi)}$$

where:

- A is the amplitude of the signal,
- $\theta[n]$ is the phase corresponding to the transmitted symbol (taking values from $\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$ for QPSK),
- ϕ is the true phase offset.

2. Received Signal with Noise

The received signal, including noise, is given by:

$$r[n] = s[n] \cdot e^{j\phi} + w[n]$$

where:

- $r[n]$ is the received signal,
- $w[n]$ represents the additive noise, modeled as $w[n] \sim \mathcal{CN}(0, \sigma^2)$, where $\sigma^2 = \frac{1}{2 \cdot 10^{\frac{SNR}{10}}}$ (variance of the complex Gaussian noise).

3. Maximum Likelihood (ML) Carrier Phase Estimation

The ML estimate of the phase can be computed as:

$$\hat{\phi}_{ML} = \arg \left(\sum_{n=1}^N r[n] \cdot e^{-j\theta[n]} \right)$$

where:

- $\hat{\phi}_{ML}$ is the estimated phase,
- N is the number of received symbols,
- $\arg(\cdot)$ is the argument (phase) function.

4. Phase-Locked Loop (PLL)

The PLL iteratively estimates the phase with the following equations:

Phase Error Calculation:

$$\text{Phase Error}[n] = \arg \left(r[n] \cdot e^{-j\hat{\phi}[n-1]} \right)$$

where:

- $\hat{\phi}[n-1]$ is the estimated phase from the previous sample.

Update Rule:

$$\hat{\phi}[n] = \hat{\phi}[n-1] + K \cdot \text{Phase Error}[n]$$

where:

- K is the loop gain or bandwidth (loop bandwidth).

5. Corrected Received Signal

The corrected received signal after applying phase estimation is:

$$r_{corrected}[n] = r[n] \cdot e^{-j\hat{\phi}[n]}$$

6. Effects of Additive Noise on Phase Estimation

The variance of the phase estimation error can be calculated as:

$$\text{Var}(\text{Phase Error}) = \text{E} [(\angle r[n] - \phi)^2]$$

where:

- $\angle r[n]$ is the phase of the received signal,
- ϕ is the true phase.

7. Decision-Directed and Non-Decision-Directed Loops

Decision-Directed Phase Estimation:

- The detected symbol is obtained from:

$$\hat{s}_{DD}[n] = e^{j\left(\frac{2\pi}{M} \cdot \text{round}\left(\frac{\angle r[n]}{2\pi} \cdot M\right)\right)}$$

- The phase error is updated similarly:

$$\text{Phase Error}_{DD}[n] = \arg \left(\hat{s}_{DD}[n] \cdot e^{-j\hat{\phi}_{DD}[n-1]} \right)$$

- Update rule:

$$\hat{\phi}_{DD}[n] = \hat{\phi}_{DD}[n-1] + K \cdot \text{Phase Error}_{DD}[n]$$

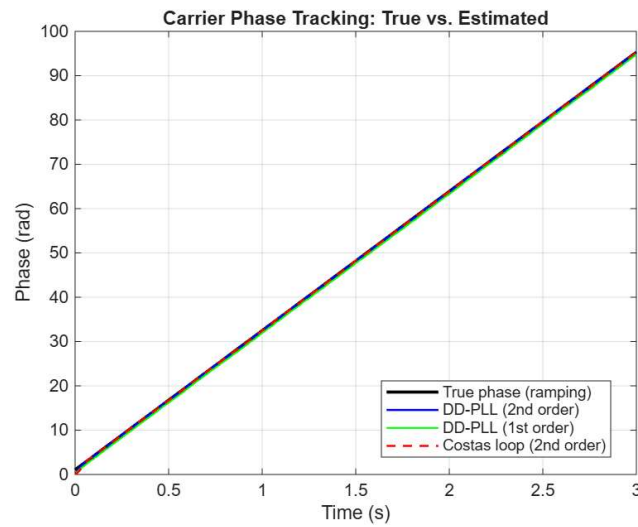
Non-Decision-Directed Phase Estimation:

- Phase error calculation is the same as in the PLL:

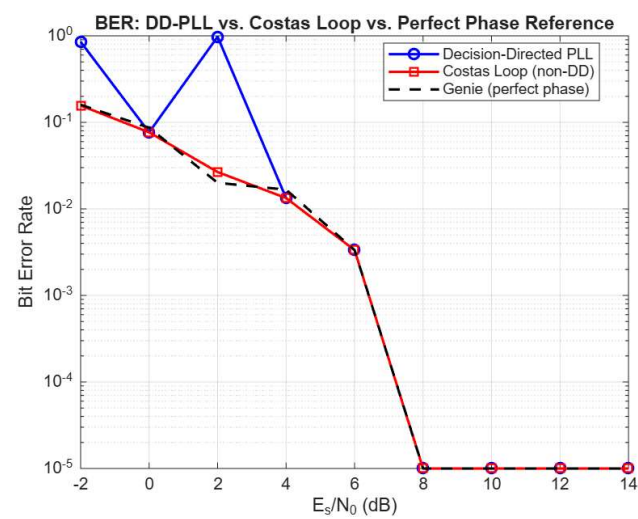
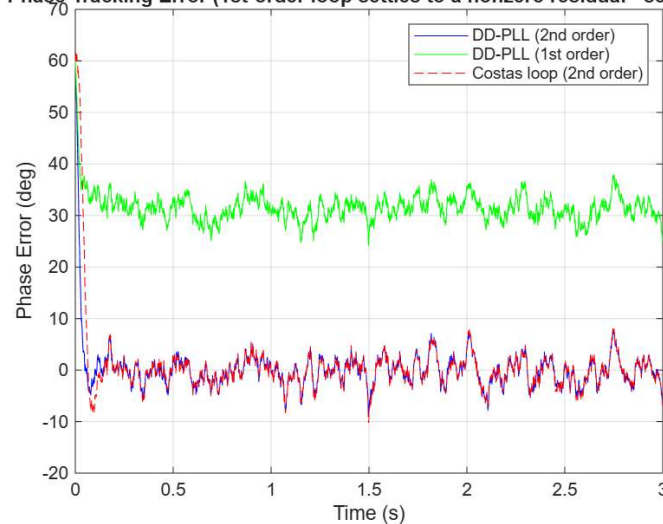
$$\text{Phase Error}_{NDD}[n] = \arg \left(r[n] \cdot e^{-j\hat{\phi}_{NDD}[n-1]} \right)$$

- Update rule:

$$\hat{\phi}_{NDD}[n] = \hat{\phi}_{NDD}[n-1] + K \cdot \text{Phase Error}_{NDD}[n]$$



Phase Tracking Error (1st-order loop settles to a nonzero residual - see note)



Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS			
Trial	Loop Type (DD/NDD)	Phase Offset (deg)	SNR (dB)	Estimated Phase (deg)	Phase Error (deg)	Constellation (before/after correction)	Convergence (fast/slow)
1	Decision-Directed	15	15				
2	Decision-Directed	45	15				
3	Decision-Directed	45	5				
4	Non-Decision-Directed	15	15				
5	Non-Decision-Directed	45	15				
6	Non-Decision-Directed	45	5				

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT - 9

M-ary Phase Shift Keying Modulation and Demodulation using USRP 2920 Software Defined Radio

Aim: To implement M-ary PSK transceiver using USRP 2920 SDR kit.

Tools required: Desktop or Laptop with labview software and USRP 2920 SDR kits.

Theory: Phase shift keying (PSK) is a one of most frequently efficient modulation technique for wireless data transmission for its high noise immunity and narrow frequency bandwidth. Binary phase shift keying (BPSK) is the simplest type of phase keying. It is normally used for high speed data transfer application, provides a 3dB power advantage over the BASK modulation technique and is robust and simple in implementation but proves to be an inefficient user of the provided bandwidth and is normally termed as a non-linear modulation scheme. Fig9.1 shows the block diagram for BPSK transmitter and receiver.

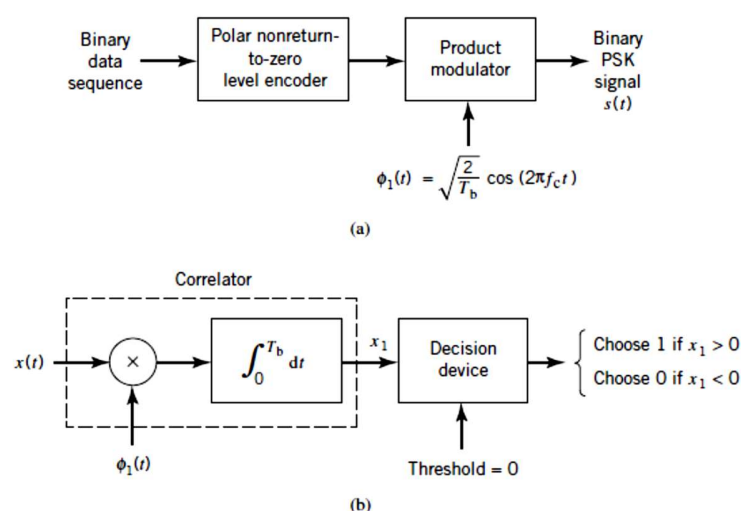


Fig9.1: BPSK (a) Transmitter (b) Receiver

In QPSK, the data bits to be modulated are grouped into symbols, each containing two bits, and each symbol can take on one of four possible values: 00, 01, 10, or 11. During each symbol interval, the modulator shifts the carrier to one of four possible phases corresponding to the four possible values of the input symbol. In the ideal case, the phases are each 90 degrees apart, and these phases are usually selected such that the signal constellation matches the configuration shown in Figure 9.2. and Figure 9.3 shows the message and corresponding QPSK signal.

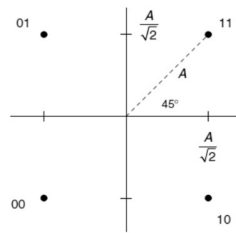


Fig9.2 : QPSK constellation.

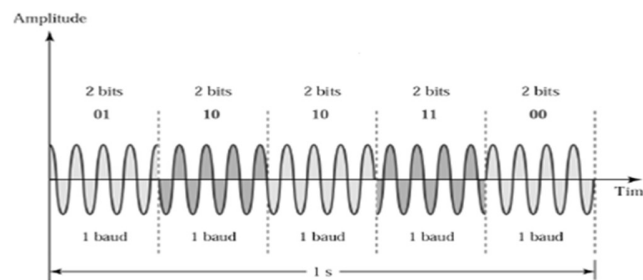


Fig 9.3: QPSK signal and the message

Procedure:

1. The block diagram and front panel for MPSK transmitter with Labview is shown in fig 9.4 and 9.5 respectively. Vary the input parameters and observe the results to understand the significance and applications of MPSK.
2. The block diagram and front panel for MPSK receiver with Labview is shown in fig 9.6 and 9.7 respectively. Vary the input parameters and observe the results to understand the significance and applications of MPSK.

The path for the sub vi's of PSK is described here:

Search where LabVIEW software is installed.

TRANSMITTER

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2. **mod_Continuous BB-PSK generation.vi: Not available in Search**
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RECIEVER

4. **MT Generate PSK System Parameters.vi: Not available in Search**

Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General

5. MT Generate Filter Coefficients.vi: Available in search or Quick drop (ctrl + Spacebar)

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6. MT Resample(Complex Cluster).vi: Available in search or Quick drop (ctrl + Spacebar)

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7. MT Demodulate PSK.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\Demodulation

8. subTruncate FilterCplxClster.vi: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments\Labview2020\examples\instr\ModulationToolkitExamples\subVIs\

9. MT Format Constellation: Available in search or Quick drop (ctrl + Spacebar)

Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General

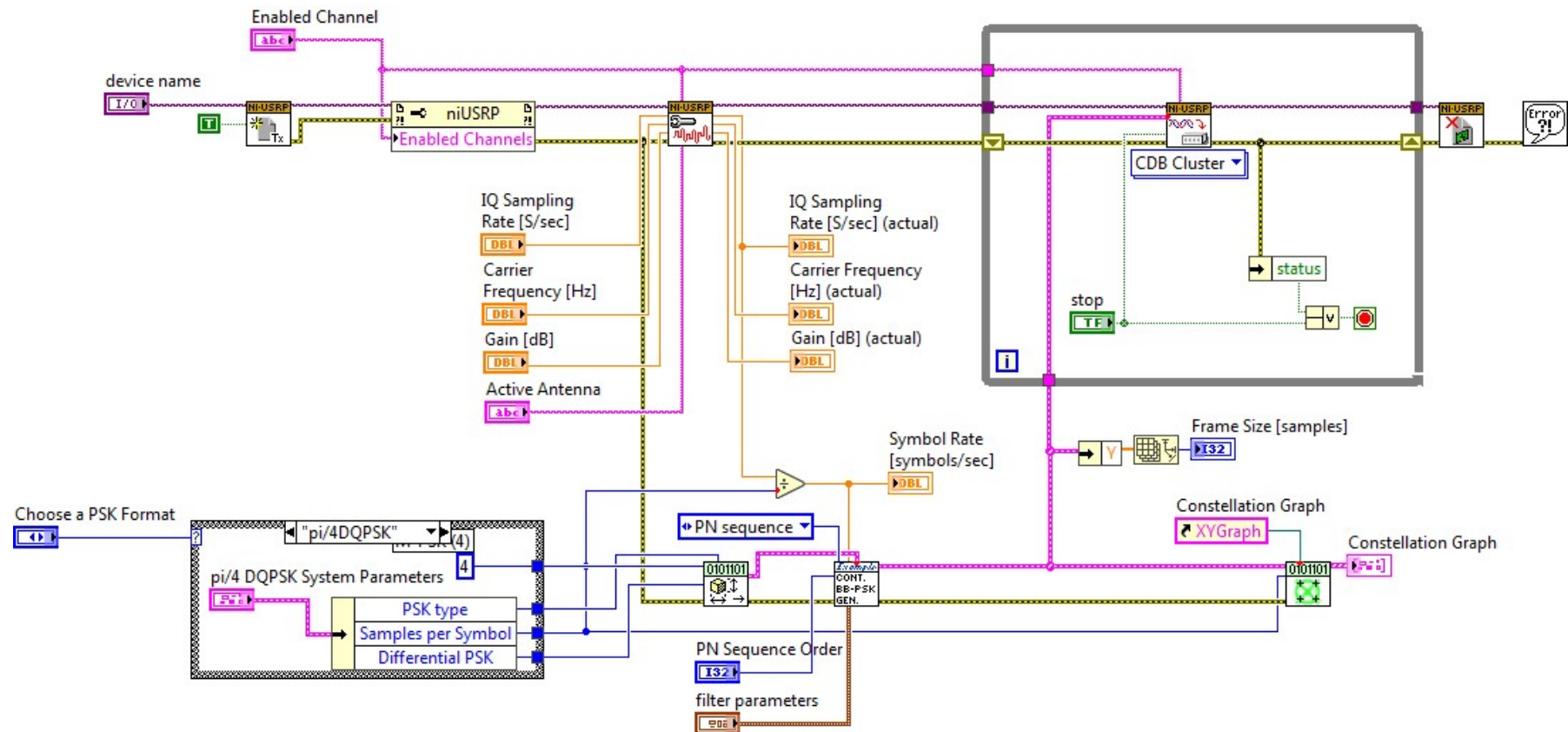


Fig9.4: Block diagram of MPSK transmitter using USRP

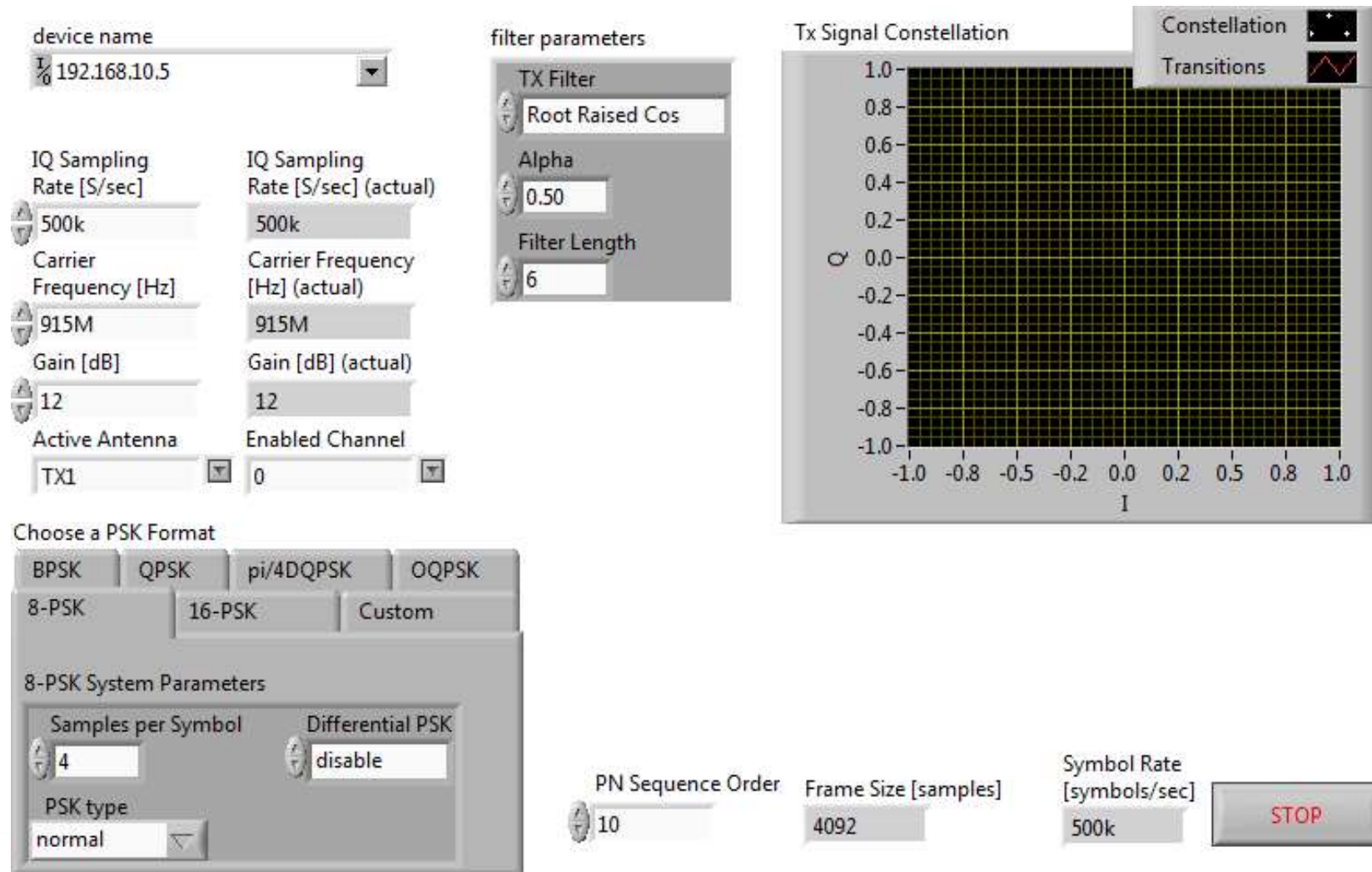


Fig9.5: Front panel of MPSK transmitter using USRP

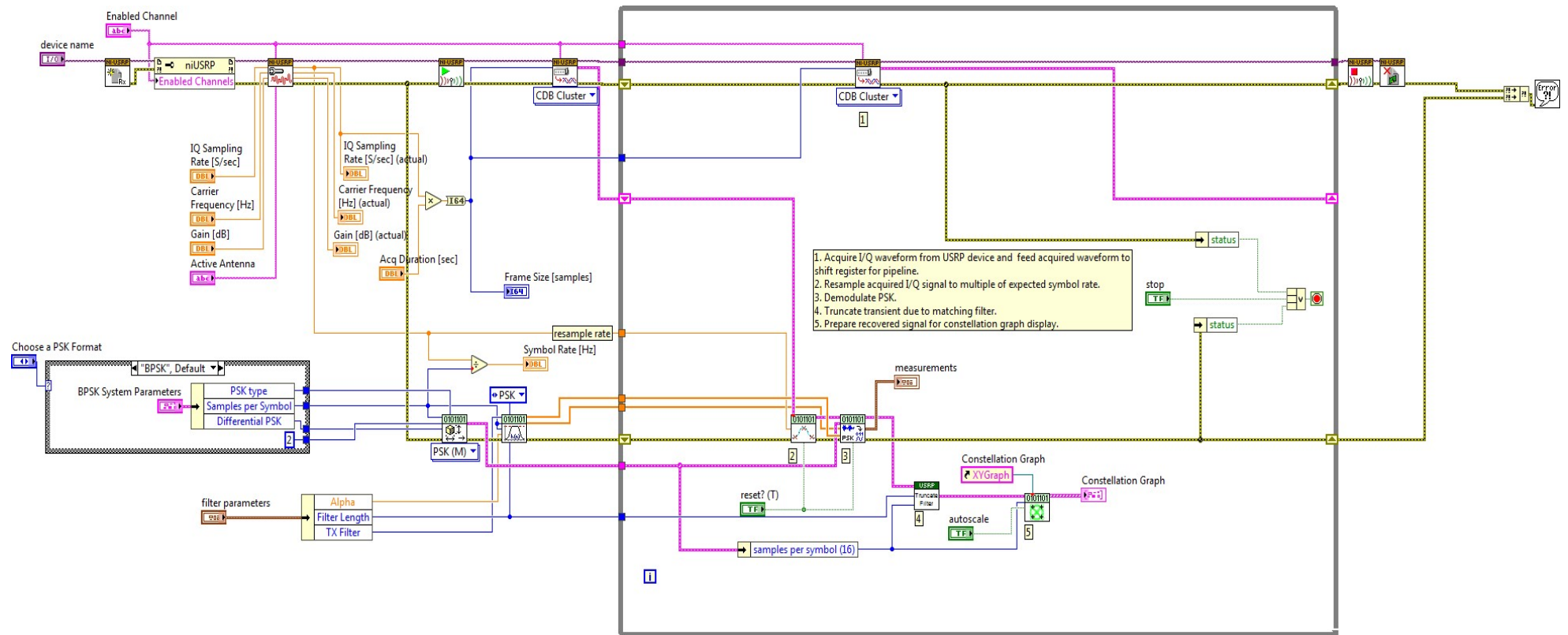


Fig9.6: Block diagram of MPSK receiver using USRP

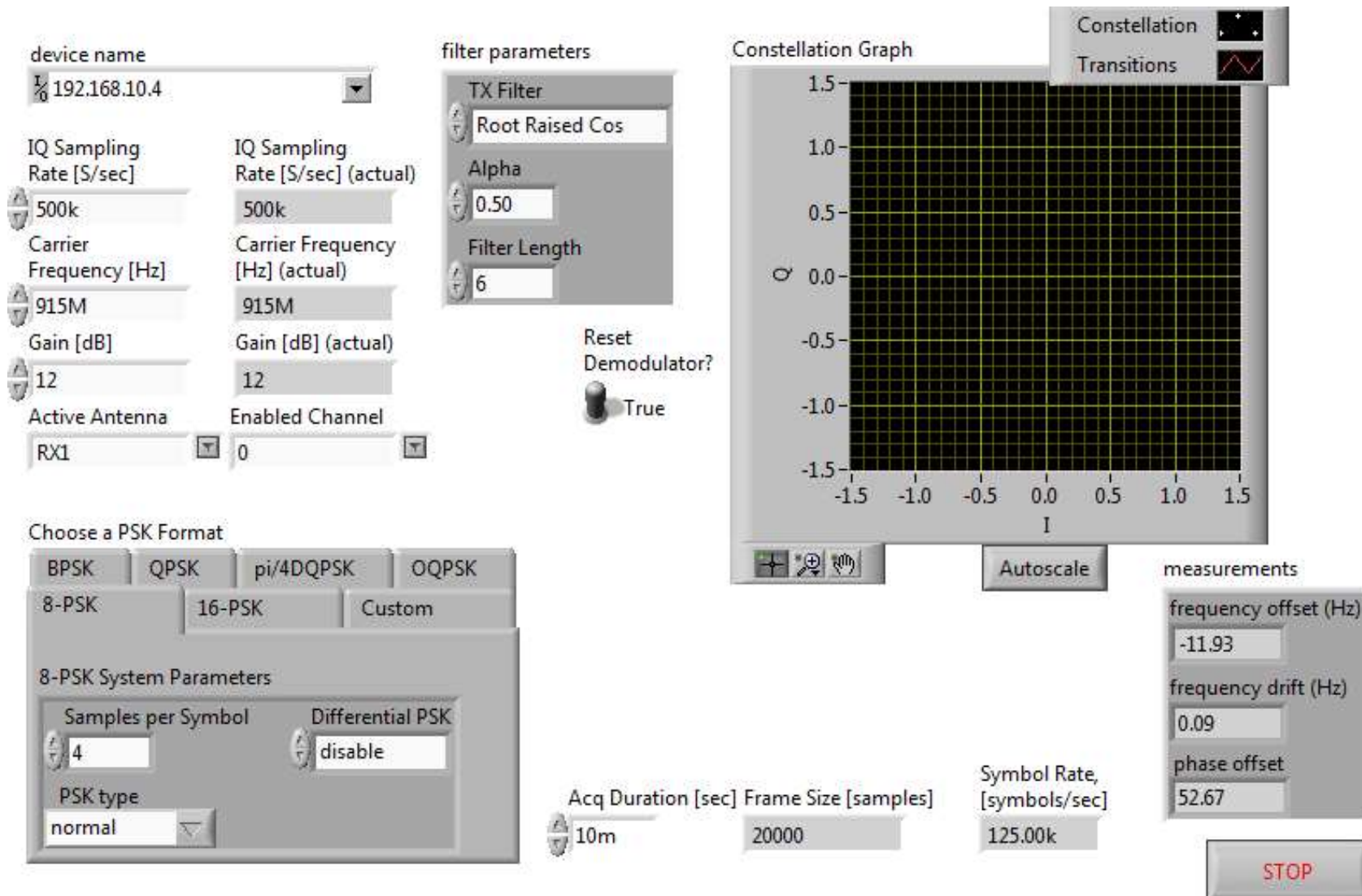


Fig9.7: Front panel of MPSK receiver using USRP

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS			
Trial	M (No. of phases)	Carrier Frequency (MHz)	Sample Rate (MSps)	Tx Spectrum	Rx Constellation	Received Symbols (match Tx?)	BER
1	2 (BPSK)	915	1				
2	4 (QPSK)	915	1				
3	8 (8-PSK)	915	1				
4	16 (16-PSK)	915	1				
5	4 (QPSK)	2400	1				
6	4 (QPSK)	915	2				

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT - 10

M-ary QAM Modulation and Demodulation using USRP 2920 Software Defined Radio

Aim: To implement M-ary QAM transceiver using NI-USRP2920 kit

Tools required: Desktop or Laptop with labview software and USRP 2920 SDR kits.

Theory: In an M-ary PSK system, the in-phase and quadrature components of the modulated signal are interrelated in such a way that the *envelope is constrained to remain constant*. This constraint manifests itself in a circular constellation for the message points. However, if this constraint is removed so as to permit the in-phase and quadrature components to be independent, we get a new modulation scheme called *M-ary QAM*. The QAM is a *hybrid* form of modulation, in that the carrier experiences amplitude as well as phase-modulation. In *M-ary PAM*, the signal-space diagram is one-dimensional. *M-ary QAM* is a two-dimensional generalization of *M-ary PAM*. The constellation diagram for QAM with $M=16$ is shown in fig 10.1.

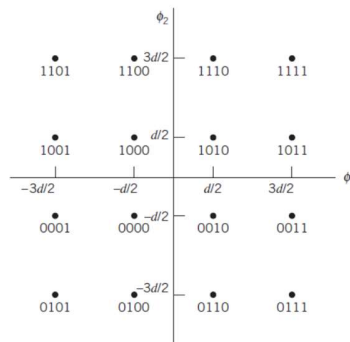


Fig 10.1: Signal-space diagram of *M-ary QAM* for $M = 16$

Procedure:

1. The block diagram and front panel for MQAM transmitter with Labview is shown in fig 10.2 and 10.3 respectively. Vary the input parameters and observe the results to understand the significance and applications of MQAM.
2. The block diagram and front panel for MQAM receiver with Labview is shown in fig 10.4 and 10.5 respectively. Vary the input parameters and observe the results to understand the significance and applications of MQAM.

The path for the sub vi's of QAM is described here:

Search where LabVIEW software is installed.

TRANSMITTER

1. **subConvertQAMM-AryEnumtoNum.vi: Available in search or Quick drop (ctrl + Spacebar)**
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2. **MT Generate System Parameters.vi: Not available in Search**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General
3. **MT Format Constellation.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General
4. **mod_Continuous BB-QAM generation.vi: Not available in Search**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\Support\examples

RECEIVER

5. **subConvertQAMM-AryEnumtoNum.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\examples\instr\ModulationToolkitExamples\subVIs\
6. **MT Generate System Parameters.vi: Not available in Search**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General
7. **MT Generate Filter Coefficients.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General
8. **MT Resample(Complex Cluster).vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Analog
9. **MT Demodulate QAM.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\Demodulation
10. **subTruncate FilterCplxClster.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\examples\instr\ModulationToolkitExamples\subVIs\
11. **MT Measure Quadrature Impairment.vi: Not available in Search**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\Demodulation
12. **MT Format Constellation.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General

QAM modulation and demodulation implementation on NI-USRP2920 kit

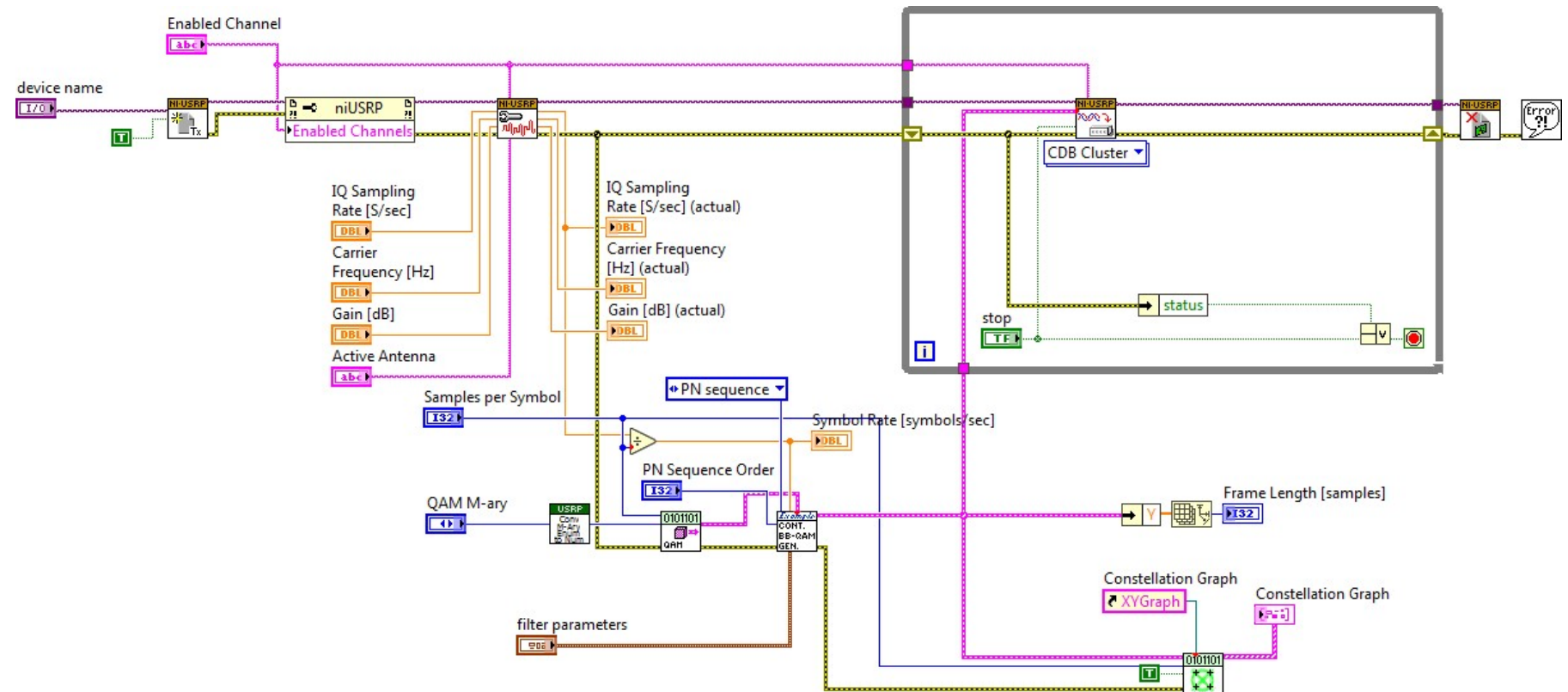


Fig10.2: Block diagram of M-QAM transmitter

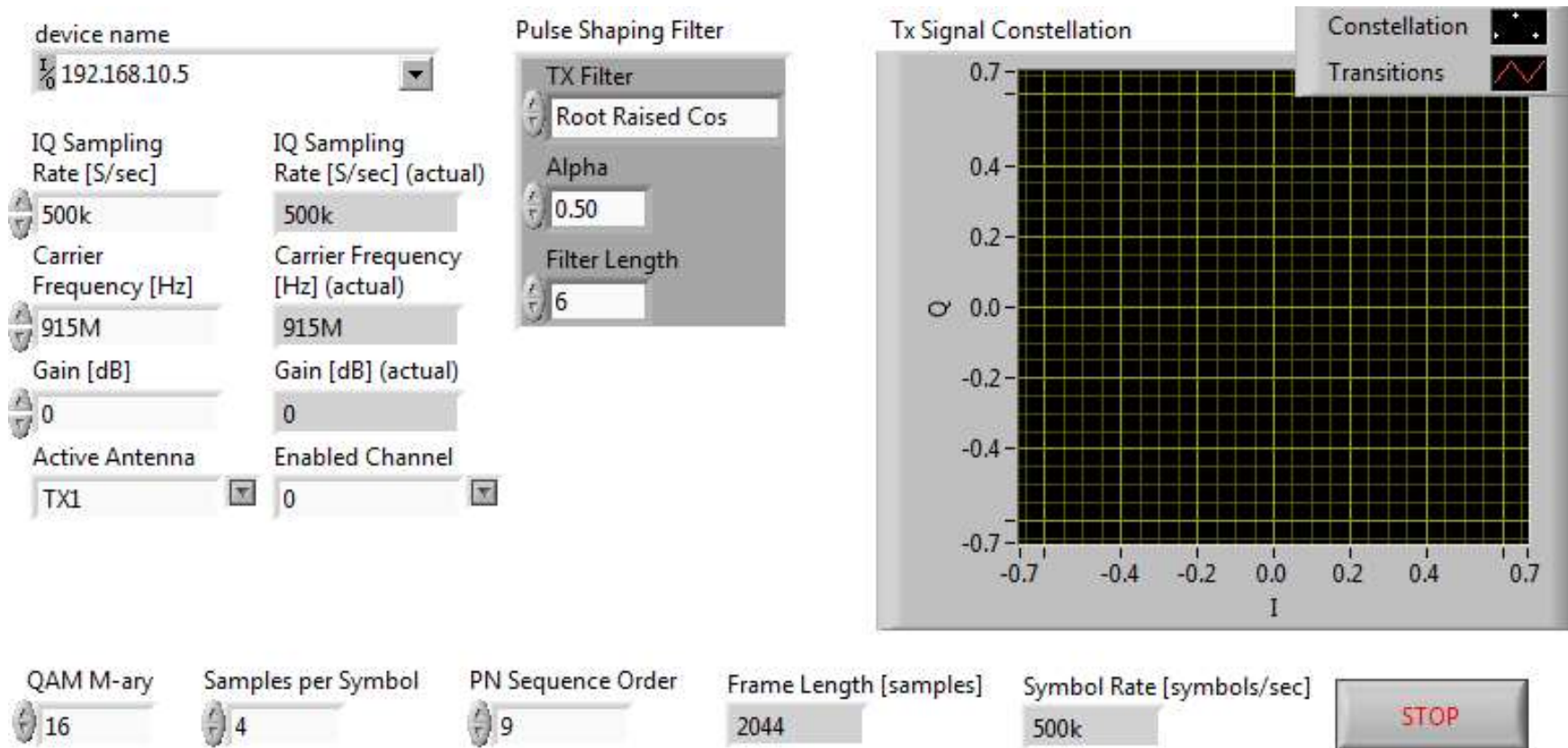


Fig10.3: Front panel of a M-QAM transmitter

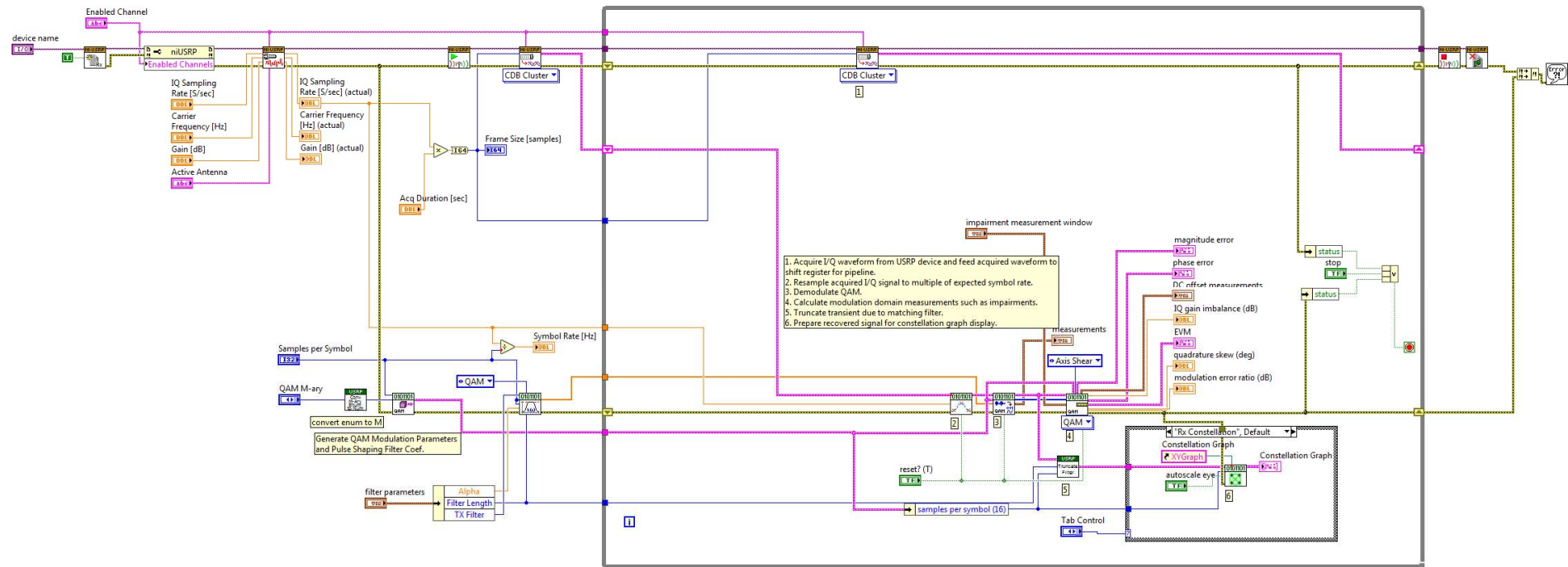


Fig10.4: Block diagram of M-QAM Rx

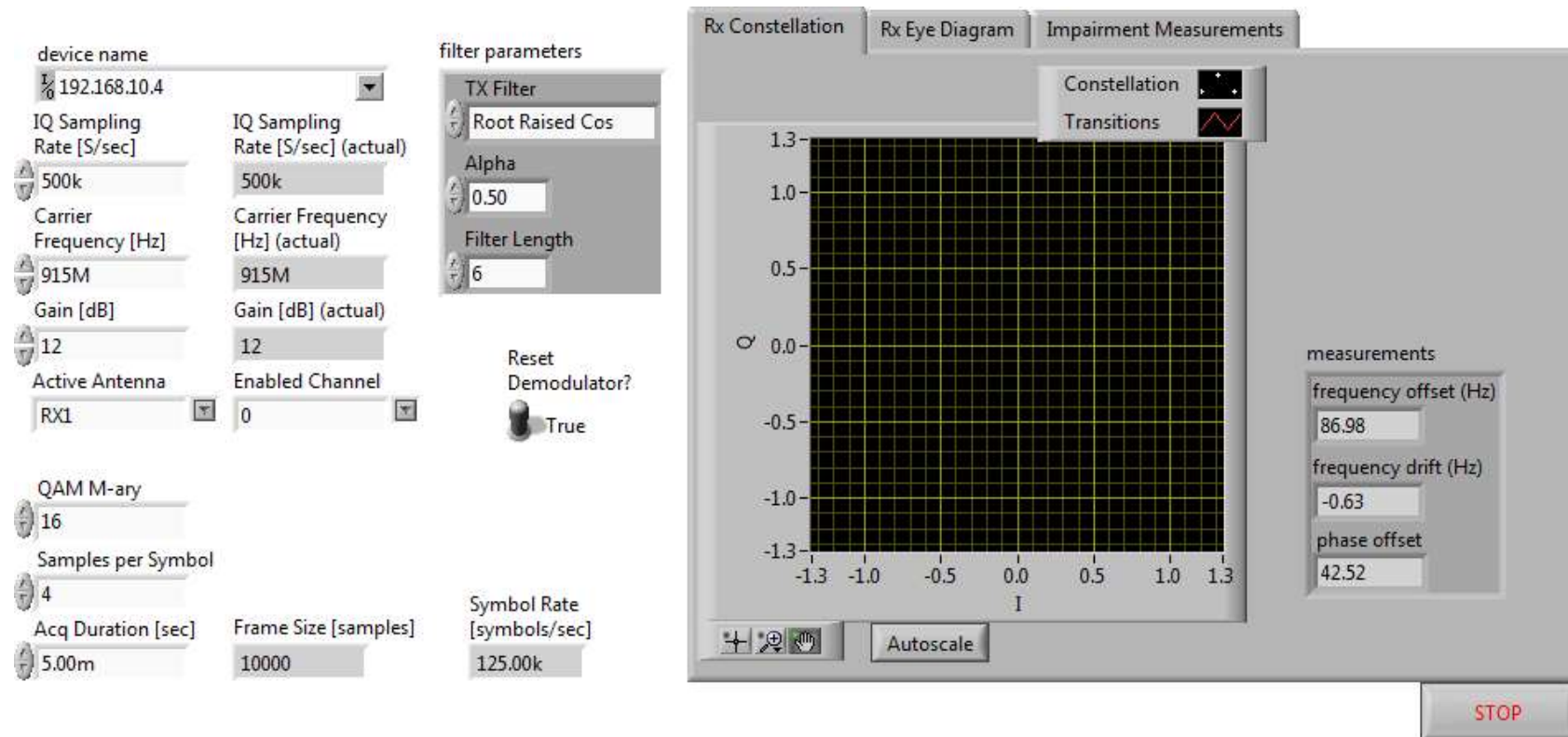


Fig10.5: Front panel of a M-QAM receiver

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS			
Trial	M (QAM order)	Carrier Frequency (MHz)	Sample Rate (MSps)	Tx Spectrum	Rx Constellation	BER	Modulation Error Ratio (MER)
1	4	915	1				
2	16	915	1				
3	64	915	1				
4	16	2400	1				
5	16	915	2				
6	64	915	2				

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT - 11

M-ary Frequency Shift Keying Generation and Detection

Aim: FSK modulation and demodulation of MFSK modulation and demodulation using USRP 2920 kit

Tools required: Desktop or Laptop with labview software and USRP 2920 SDR kits.

Theory: Frequency shift keying (FSK) is the most common form of digital modulation in the high-frequency radio spectrum and has important applications in telephone circuits. Binary FSK is a modulation scheme typically used to send digital information between digital equipment such as teleprinters and computers. The data are transmitted by shifting the frequency of a continuous carrier in a binary manner to one or the other of two discrete frequencies. One frequency is designated as the “mark” frequency and the other as the “space” frequency. The mark and space correspond to binary one and zero, respectively. By convention, mark corresponds to the higher radio frequency. Figure 11.1 shows the relationship between the data and the transmitted signal.

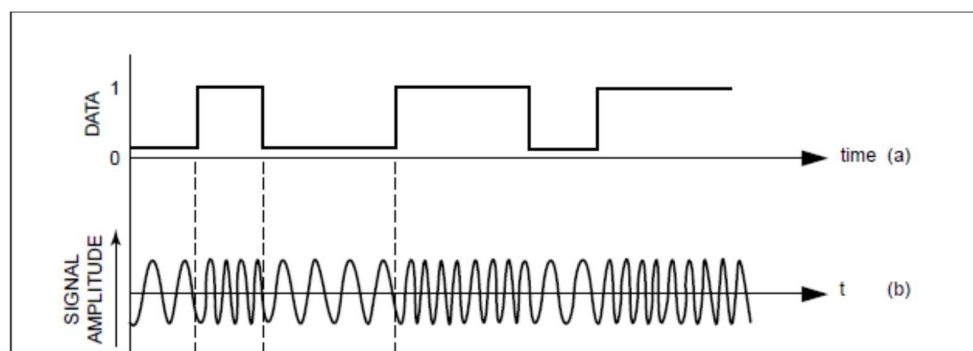


Fig11.1: FSK modulation. (a) Binary data (b) frequency modulates the carrier to produce the FSK signal

Procedure:

1. The block diagram and front panel for MFSK transmitter with Labview is shown in fig 11.2 and 11.3 respectively. Vary the input parameters and observe the results to understand the significance and applications of MFSK.
2. The block diagram and front panel for MFSK receiver with Labview is shown in fig 11.4 and 11.5 respectively. Vary the input parameters and observe the results to understand the significance and applications of MFSK.

The path for the sub vi's of MFSK is described here:

Search where LabVIEW software is installed.

TRANSMITTER

1. **MT Generate FSK System Parameters.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\General
2. **mod_Create Continous FSK.vi: Not available in Search**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\Support\examples

RECEIVER

3. **MT Generate Filter Coefficients.vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\LabVIEW 2020\vi.lib\addons\Modulation\Digital\General
4. **mod_FSK Deviation Measurements.vi: Not available in Search**
Path: National Instruments\LabVIEW 2020\vi.lib\addons\Modulation\Digital\Support\examples
5. **mod_Plot FSK Eye Diagram.vi: Not available in Search**
Path: National Instruments\LabVIEW 2020\vi.lib\addons\Modulation\Digital\Support\examples
6. **MT Resample(Complex Cluster).vi: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Analog
7. **MT Demodulate FSK: Available in search or Quick drop (ctrl + Spacebar)**
Path: National Instruments\Labview2020\vi.lib\addons\Modulation\Digital\Demodulation

device name
192.168.10.2

Tx IQ Sampling Rate [S/sec]
200k

Tx Frequency [Hz]
915M

Tx Gain [dB]
0

Active Antenna
TX1

Tx IQ Sampling Rate [S/sec] (actual)
0

Tx Frequency [Hz] (actual)
0

Tx Gain [dB] (actual)
0

FSK Parameters

PN Sequence Order
9

FSK M-ary
2

Samples Per Symbol
4

Specify Modulation Index

Specify FSK Deviation

FSK Deviation
50k

Pulse Shaping Filter Parameters

TX Filter
Gaussian

Alpha
0.50

Filter Length
4

NOT GENERATING

Frequency Deviation [Hz]
50k

Symbol Rate [Hz]
44.44k

Signal Bandwidth [Hz]
166.67k

Error Out

status	code
	d0

source

STOP

Fig11.3: Front panel of MFSK Tx

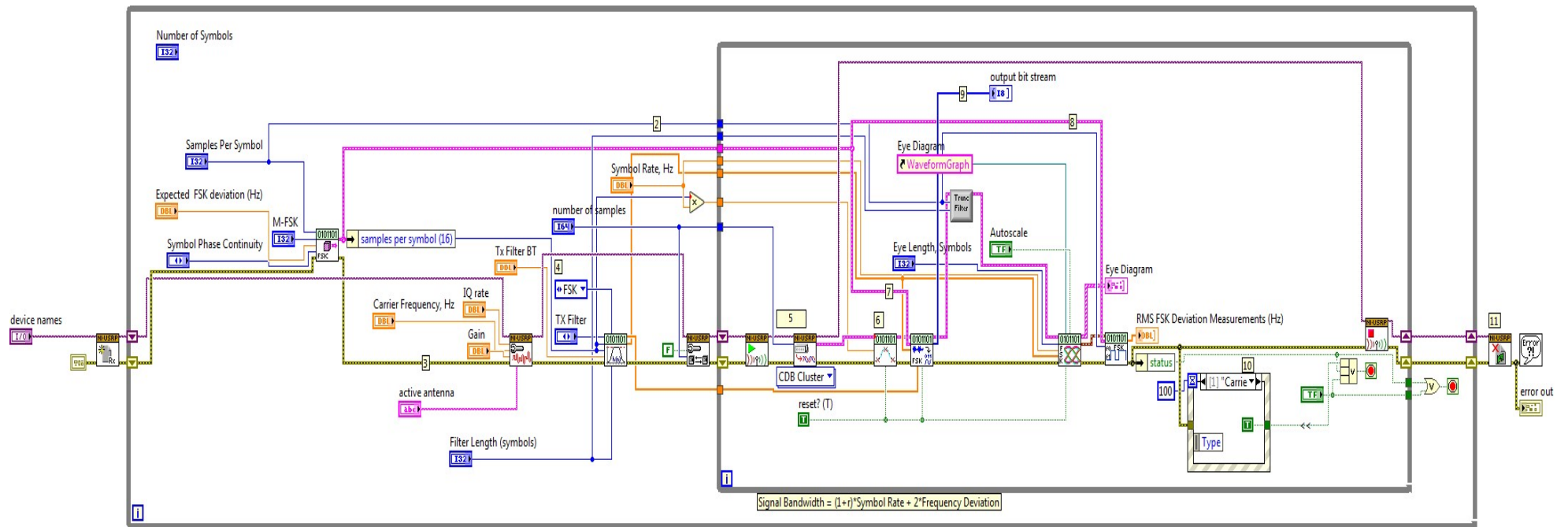


Fig11.4: Block diagram of MFSK Rx

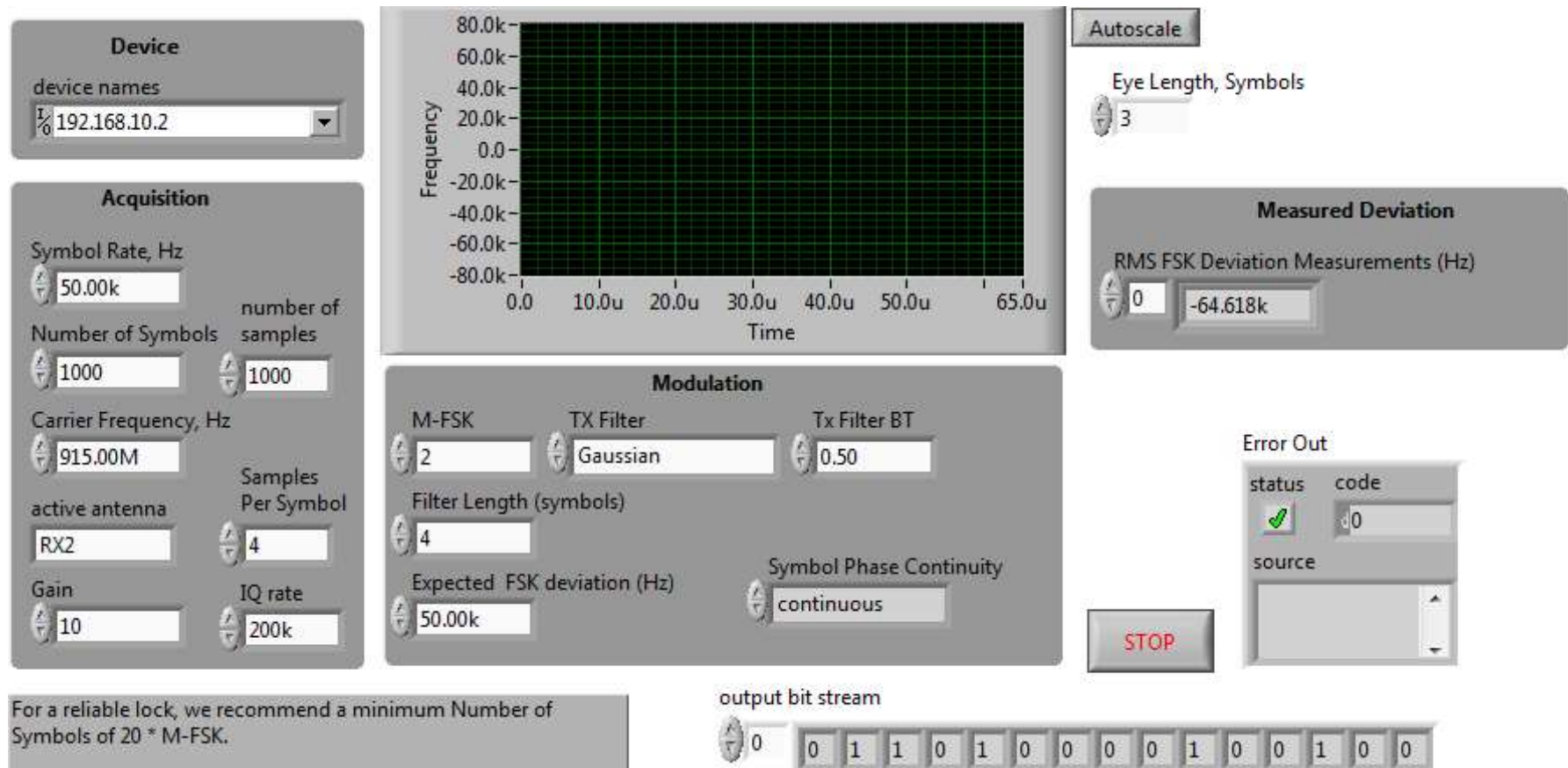


Fig11.5: Front panel of MFSK Rx

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS		
Trial	M (No. of frequencies)	Frequency Deviation (Hz)	Carrier Frequency (MHz)	Eye Diagram (Rx)	Measured Frequency Deviation (Hz)	BER
1	2 (BFSK)	1000	915			
2	4	1000	915			
3	8	1000	915			
4	4	2000	915			
5	4	500	915			
6	4	1000	2400			

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

EXPERIMENT 12

Adaptive Equalization: Zero-Forcing (ZFE), MSE/LMS and Decision-Feedback Equalizer (DFE)

Aim: To design, simulate, and compare adaptive Zero-Forcing (ZF), Minimum Mean Square Error (MMSE/LMS), and Decision-Feedback (DFE) equalizers for combating intersymbol interference (ISI) in a dispersive baseband digital communication channel using MATLAB

Tools required: Desktop or Laptop with MATLAB software.

Theory:

In a band-limited or multipath dispersive channel, successive transmitted symbols spread into adjacent symbol intervals, producing Intersymbol Interference (ISI). An adaptive equalizer is a filter placed at the receiver that is adjusted, using a known training sequence and/or decision-directed feedback, to compensate for the channel distortion and recover the original symbols. The discrete-time baseband channel model used in this experiment is:

$$r(n) = \sum_{k=0}^{L-1} h(k) a(n-k) + w(n)$$

where $h(k)$, $k = 0, \dots, L-1$ are the channel taps, $a(n)$ is the transmitted BPSK symbol sequence, and $w(n)$ is additive white Gaussian noise of variance σ^2 . A linear transversal equalizer with N_f taps produces an estimate of the transmitted symbol as:

$$y(n) = \sum_{k=0}^{N_f-1} c_k r(n-k)$$

The optimum tap-weight vector c is found by solving the discrete Wiener-Hopf (normal) equations, $R \cdot c = p$, where R is the correlation matrix of the received sequence and p is the cross-correlation vector between the desired symbol $a(n-D)$ and the received samples. Both quantities are built from the channel autocorrelation function:

$$\Phi(k, j) = \sum_{i=0}^{L-1} h(i) h(i+k-j)$$

$$p(k) = E_s h(D-k)$$

D is the overall decision delay chosen so the equalizer output best aligns with the dominant channel tap. The channel itself is not known a priori; it is estimated from the training sequence using a least-squares fit before either equalizer is designed, so both filters below are genuinely data/adaptive-derived.

Zero-Forcing Equalizer (ZFE)

The Zero-Forcing equalizer is designed to force the combined channel-equalizer response to a single impulse (zero ISI), completely ignoring the effect of noise. Its coefficients solve the normal equations built from the noise-free correlation matrix:

$$\mathbf{R}_{zf} = E_s \Phi, \quad \mathbf{c}_{zf} = \mathbf{R}_{zf}^+ \mathbf{p}$$

Because noise is excluded from the design, the ZF solution can require very large tap gains near a spectral null of the channel, which amplifies the noise at the equalizer output. This effect, known as noise enhancement, is the fundamental weakness of the Zero-Forcing criterion.

2. Minimum Mean Square Error (MMSE) / LMS Adaptive Equalizer

The MMSE equalizer minimizes the mean squared error between the transmitted symbol and the equalizer output, explicitly including the noise variance σ^2 in its design equation:

$$\mathbf{R}_{mmse} = E_s \Phi + \sigma^2 \mathbf{I}, \quad \mathbf{c}_{mmse} = \mathbf{R}_{mmse}^{-1} \mathbf{p}$$

This regularization keeps the tap-weight energy bounded even near a spectral null, trading a small amount of residual ISI for substantially better noise performance than the ZF solution. In practice the MMSE solution is reached adaptively, symbol-by-symbol, using the Least-Mean-Squares (LMS) stochastic-gradient algorithm, which requires no matrix inversion. Given the instantaneous error

$$e(n) = d(n) - y(n)$$

the tap weights are updated after every received symbol as:

$$c_k(n+1) = c_k(n) + \mu e(n) r(n-k)$$

where μ is the adaptation step-size. During an initial training phase $d(n)$ is a known transmitted symbol; afterwards the equalizer switches to decision-directed mode, where $d(n)$ is replaced by the receiver's own hard decision, $\text{sign}[y(n)]$.

3. Decision-Feedback Equalizer (DFE)

A DFE augments the linear feed-forward filter with a feedback filter driven by past DECISIONS (assumed correct) to cancel the trailing (post-cursor) ISI directly, without needing extra feed-forward gain and hence without the associated noise enhancement:

$$y(n) = \mathbf{c}^T \mathbf{r}(n) + \mathbf{b}^T \hat{\mathbf{a}}(n)$$

Both filters are updated by the same LMS rule, driven by a common error signal:

$$c_k(n+1) = c_k(n) + \mu e(n) r(n-k), \quad b_j(n+1) = b_j(n) + \mu e(n) \hat{a}(n-D-j)$$

Because the feedback path relies on correct past decisions, an incorrect decision can propagate into subsequent symbol estimates ("error propagation"). This makes the DFE most advantageous at moderate-to-high SNR, where decisions are usually reliable; at very low SNR its performance can fall below that of a purely linear equalizer.

4. Performance Metrics

In addition to the Bit Error Rate (BER), the following standard metrics were used to characterize and compare the three equalizers. The residual ISI (peak distortion) of the combined channel-equalizer response $g = h * c$ is:

$$D_{ISI} = \frac{1}{|a(0)|} \sum |g(n)|, \quad g = h * c$$

The output SNR, which directly captures the noise-enhancement penalty, is:

$$SNR_{out} = \frac{|g(0)|^2}{\sigma^2 \sum c^2}$$

The theoretical minimum (Wiener) MSE floor and the LMS misadjustment (its steady-state excess MSE, normalized) are:

$$J_{min} = E_s - \mathbf{p}^T \mathbf{c}_{mmse}$$

$$M = \frac{J_{excess}}{J_{min}}, \quad J_{excess} \approx \frac{\mu}{2} \text{tr}(\mathbf{R}) J_{min}$$

Output:

The channel used was the 3-tap raised-cosine test channel $h(n)=0.5[1+\cos(2\pi/W(n-2))]$ with $W = 2.9$ (a near-spectral-null, high-eigenvalue-spread channel), $SNR = 12$ dB, $N_f = 15$ feed-forward taps and $N_b = 5$ feedback taps. The following figures show the combined channel-equalizer response, the LMS/DFE learning curves, the BER comparison at 12 dB, and a BER-vs-SNR sweep.

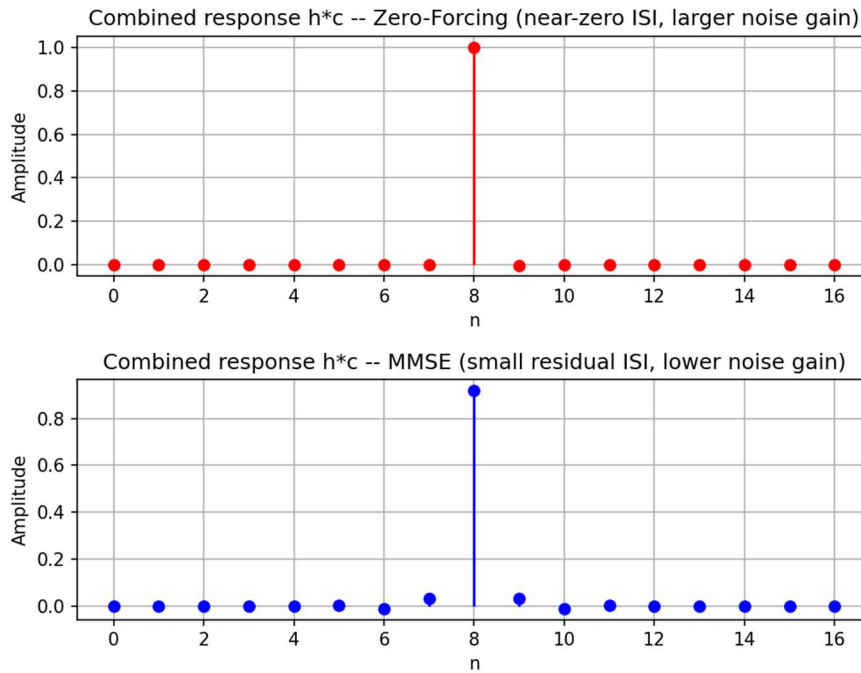


Fig 12.1 - Combined channel-equalizer response $h*c$: Zero-Forcing (top, exactly one non-zero tap) vs. MMSE (bottom, small residual ISI).

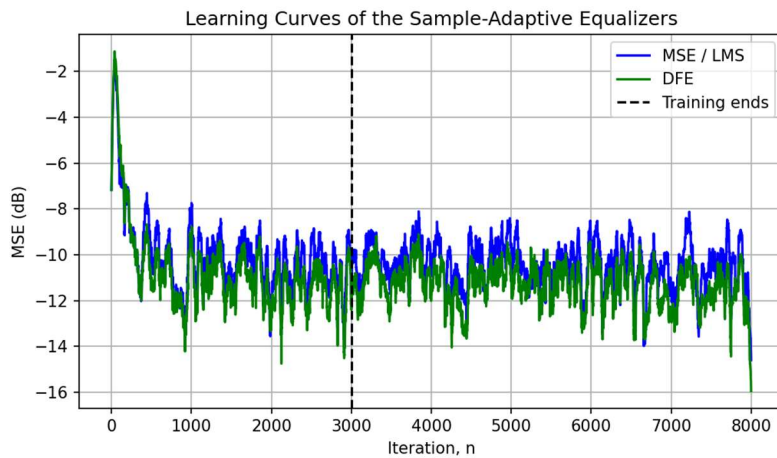


Fig 12.2 - MSE learning curves of the sample-adaptive LMS and DFE equalizers.

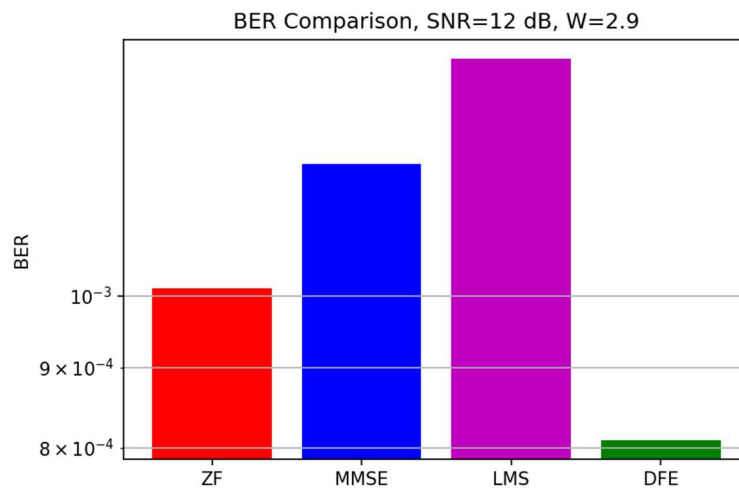


Fig 12.3 - BER comparison of ZF, MMSE, LMS and DFE equalizers at SNR = 12 dB.

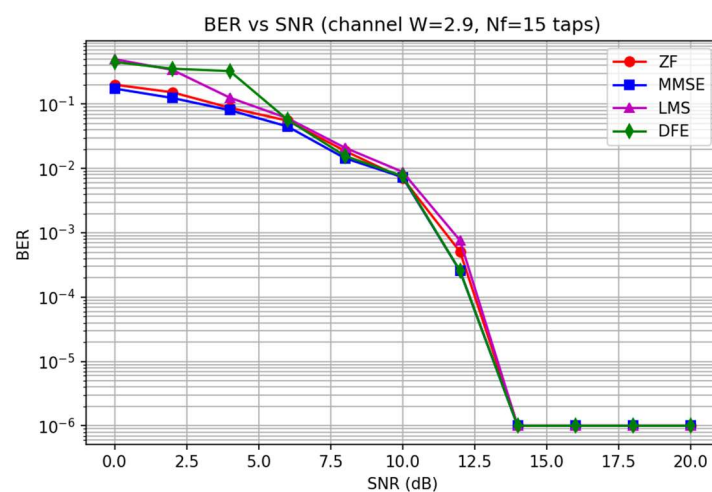


Fig 12.4 - BER vs SNR sweep (0-20 dB) for all four equalizers.

Conclusion:

INPUT PARAMETERS				OUTPUT PARAMETERS			
Trial	Equalizer Type (ZF/MMSE-LMS/DFE)	SNR (dB)	Step size (μ) / Filter length	Residual ISI (D_ISI)	Output SNR (dB)	MSE Floor / Misadjustment	BER
1	ZF	12	-- / 11				
2	MMSE/LMS	12	0.01 / 11				
3	DFE	12	0.01 / 11+5				
4	MMSE/LMS	0	0.01 / 11				
5	MMSE/LMS	20	0.01 / 11				
6	DFE	20	0.01 / 11+5				

Sl.No	Criteria	Max Marks	Marks obtained
Data sheet			
A	Problem statement	10	
B	Design & specifications	10	
C	Expected output	10	
Record			
D	Simulation/ Conduction of the experiment	15	
E	Analysis of the result.	15	
	Viva	40	
	Total	100	
Scale down to 10 marks			
Staff Signature with Date			

Appendix: Labview Procedure for MPSK and M-QAM

To build an MPSK transceiver, the required sub vi's and their paths in the Labview directory are illustrated in Table1. It is assumed that the Labview software is installed in the path- C:\Program files\ National Instruments\Labview2020. The sub vi's of Table1 are inside this directory. For example, to find the **mod_psk transmitter.vi**, one can use the path as C:\Program files\National Instruments\ Labview2020\ vi.lib\ addons\ Modulation\ Digital\ Modulation. These vi's are used sequentially to construct a block diagram of the M-PSK transceiver as shown in Figure 1. The vi's and their paths required to construct M-QAM are illustrated in Table2. The impairments of a communication system like AWGN (Additive White Gaussian Noise) and IQ(In phase - Quadrature phase) impairments are considered for determining BER, estimating SNR at receiver, estimating demodulated symbols, plotting the transmit spectrum, receive spectrum, constellation diagram and eye diagram.

Table 1: Labview sub-vi paths for MPSK transceiver

Sub vi's	Path
Main Blocks	
mod_psk transmitter.vi	vi.lib\addons\Modulation\Digital\Modulation
MT Add AWGN.vi	vi.lib\addons\Modulation\Digital\Impairments
MT Apply IQ Impairments.vi	
MT Demodulate PSK.vi	vi.lib\addons\Modulation\Digital\Demodulation
MT measure PSK Quadrature Impairments.vi	
MT Calculate BER.vi	
MT Calculate EbNo.vi	
Mod_align Tx and Rx bit sequence	vi.lib\addons\Modulation\Digital\support\examples
Case Structure	
Transmitted Symbols\Demodulated Symbols: MT bit stream to digital graph.vi	vi.lib\addons\Modulation\Digital\General
<i>Constellation (Tx Output): MT format Constellation.vi</i>	
<i>MT format Constellation.vi</i>	
<i>Eye Diagram(Rx): MT Format Eye Diagram.vi</i>	
<i>TX Spectrum: Mod_complex FFT(Mag_Phase).vi</i>	vi.lib\addons\Modulation\Analog\support
<i>Constellation (Rx Output): mod_truncate filter transient.vi</i>	vi.lib\addons\Modulation\Digital\support\Examples

Table 2: Labview sub-vi paths for MQAM transceiver

Sub vi's	Path
Main Blocks	
mod_QAM Transmitter.vi	vi.lib\addons\Modulation\Digital\Support\Examples
MT Add AWGN.vi	vi.lib\addons\Modulation\Digital\Impairments
MT Apply IQ Impairments.vi	
MT Demodulate QAM.vi	vi.lib\addons\Modulation\Digital\Demodulation
MT measure QAM Quadrature Impairments.vi	
MT Calculate BER.vi	
MT Calculate EbN0.vi	
MT Modulate QAM	vi.lib\addons\Modulation\Digital\Modulation
MT Measure Rho	vi.lib\addons\Modulation\Digital\Demodulation
mod_align TX and RX bit sequence	vi.lib\addons\Modulation\Digital\support\examples
Case Structure	
Transmitted Symbols\Demodulated Symbols:	vi.lib\addons\Modulation\Digital\General
MT Bit Stream to Digital Graph.vi	
Constellation (Tx Output)	
Constellation (Rx Output)	vi.lib\addons\Modulation\Digital\support\Examples
mod_Truncate Filter Transient.vi	
MT Format Constellation.vi	vi.lib\addons\Modulation\Digital\General
Eye Diagram (Rx)	
MT Format_Eye Diagram.vi	

For More Details Pl refer:

Raghunandan B H, Mahesh A, Mahesha Babu M.P, “Wireless Communication System Design Using Labview and Software Defined Radio”, 7th International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), 2023.

(Link to download: <https://ieeexplore.ieee.org/document/10334108>)

References

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